

A Safety Framework for Collaborative Human–Robot Interaction in Cable-Driven Parallel Robots

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Abstract—As Cable-Driven Parallel Robots (CDPRs) move into human-shared workspaces, human factors must be treated as a primary design objective to enable safe and usable collaboration. In practice, operators may contact a cable or the Moving-Platform (MP) during operation; although cables are traditionally considered hazards, we instead promote them to a controlled interaction channel capable of carrying both disturbances and cooperative intent. We introduce a feasibility-aware supervisory framework that unifies safety-critical and collaborative physical human-robot interaction in CDPRs using only onboard cable-tension sensing, by coordinating compliant MP motion with contacted-cable tension regulation. The central novelty is an interaction strategy supervisor that decides whether releasing a contacted cable is robustly feasible at the current MP pose before any tension is reduced, and otherwise routes the response through MP-level compliance alone. Supporting this decision leads to two new control components: an elasto-plastic compliance law that provides workflow continuity between safety reaction and intentional guidance, and a cable-tension regulation law that bounds the contacted cable below a prescribed limit while preserving wrench balance. Experiments on a CDPR platform, named [PlatformName], validate three claims: an interaction-flow scenario demonstrates unified transitions across interaction modes; repeated-task trials quantify transparency, force consistency (cross-checked against a handle-mounted F/T sensor), and non-amplifying MP-level energetics; and user trials with novices assess usability and collaborative accessibility.

Index Terms—Cable-Driven Parallel Robots, physical human–robot interaction, compliance control, wrench feasibility, collaborative robotics.

I. INTRODUCTION

CABLE-Driven Parallel Robots (CDPRs) are a class of parallel robots in which the Moving-Platform (MP) is actuated by flexible cables rather than rigid links. Replacing links with cables yields high reconfigurability, a large workspace, and a high payload-to-mass ratio, enabling high-speed motion and agile operations [1]–[3]. These properties have made CDPRs attractive in industrial and entertainment settings, and recent applications extend to medical rehabilitation, logistics, construction, and even retrofitting of residential buildings [4]–[7]. In many of these use cases, human operators work with the robot in a shared workspace, which both raises safety concerns and opens opportunities for physical Human–Robot Interaction (pHRI). This challenge is also timely from a standardisation perspective, as the 2025 revisions of ISO 10218 reaffirm the importance of safety both at the robot level and at the level of industrial robot applications and robot cells [8]. Figure 1 depicts a representative scenario in which a human operator is inside a CDPR frame, highlighting the integration

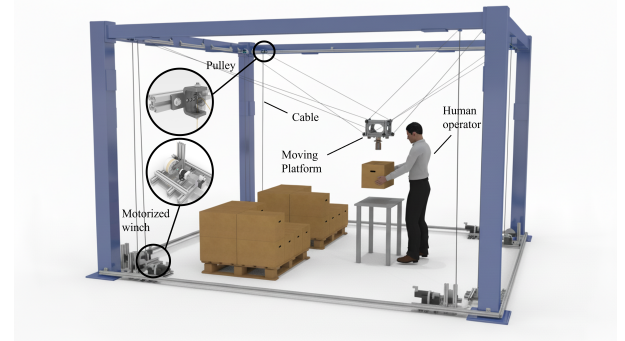


Fig. 1: Human–robot collaborative setup in which a human operator is unloading a payload delivered by a CDPR.

of human effort with robotic assistance in an industrial or logistic context.

Human–robot collaboration has been widely studied for serial and aerial robots [9]–[11]. For CDPRs, however, safety remains the dominant challenge: human–cable collisions can generate intense localised pressures, while human–MP collisions can involve substantial momentum due to the scale and dynamics of the MP [12], [13]. Prior explorations of collaboration with CDPRs have frequently adopted shared control [14], where an operator physically interacts with an instrumented end-effector interface (e.g., via a 6-axis force/torque sensor) to control the MP motion. Shared control is appropriate for teleoperation in hazardous environments because it avoids direct physical contact [15]. By contrast, direct pHRI keeps the human in the loop through physical contact with the robot, enabling more intuitive interaction without hand-held devices. Through compliance, the robot dissipates energy to improve safety and fault tolerance, while easily accommodating multiple simultaneous or sequential operators [16], [17].

Several frameworks have been proposed to address HRI with CDPRs, notably safety-oriented prototypes [18]. On the perception side, known environments can be modelled to avoid potential collisions using sampling-based planning or learning [19], [20], and geometric models of cylinders, spheres, lines, and cuboids have been developed to represent obstacles [21], [22]. In terms of the spectral content of interaction wrench signals, intentional collaboration and unintentional impacts exhibit distinct frequency profiles [23], [24]. Recent work has also begun to distinguish contacts involving different robot substructures (MP versus cables) and different task contexts [25]. On the management side, beyond emergency stops [3], a common approach is to reduce the effective stiffness of the system. In particular, reducing the tension in a collided cable mitigates impact forces and protects the operator [12], while compliance control can move

In accordance with IEEE double-blind review guidelines, all identifying information has been replaced with placeholders. The supplementary video material and supporting data are available through an anonymized link: [here](#).

the MP away after human contact [25]. Beyond compliance and tension reduction, recent work has also exploited the structural reconfigurability of CDPRs to optimise interaction performance during pHRI through online adjustment of the cable anchor positions [26], reshaping the available wrench set along the human-intent direction at the cost of actuated anchors and moving-platform force sensing. However, prior work treats a cable primarily as a hazard to avoid, and handles human–cable and human–MP contacts as separate problems: (i) cable-release methods rapidly drive the collided cable towards slackness while leaving the MP trajectory unaltered, implicitly assuming continued wrench feasibility and that the contacted tension can be reduced to zero [27]; (ii) MP-level compliance is not explicitly coupled to concurrent human–cable contacts [28]; and (iii) no existing decision layer verifies whether releasing a contacted cable is robustly feasible at the current pose before the tension is actually reduced. These limitations make it difficult to promote a cable from a mere hazard to a controlled interaction channel that carries both safety reactions and cooperative intent within a single workflow.

A second limitation is sensor economy. CDPRs span a broad range of sizes and are often very large in industrial deployments, where motion-capture precision decreases with scale and occlusions caused by people in the workspace [29], [30]. MP pose is frequently estimated rather than directly measured [31], and installing force/torque sensors on the MP is not always feasible, while human contact may occur anywhere on the MP or along any cable. Contact wrenches are therefore often inferred from cable tensions and a dynamic model [32], [33], which keeps mechatronic components away from the human-shared area but compromises the precision and stability of compliance control [16], [25], [28], [34]. A deployable pHRI framework for CDPRs must therefore rely on onboard cable-tension sensing only, without requiring direct MP force sensing or high-accuracy pose feedback.

To address these limitations, this paper proposes a *feasibility-aware supervisory framework* that unifies safety-critical and collaborative pHRI in CDPRs using onboard cable-tension measurements, by coordinating compliant MP motion with contacted-cable tension regulation. Rather than being treated solely as a hazard, a cable is promoted here to a controlled interaction channel that can convey both disturbance and cooperative intent. The same framework also manages direct interaction with the MP through compliant motion, thereby unifying human–cable and human–MP contacts. The contributions of this paper are fourfold:

- **Interaction strategy supervisor.** A decision layer that, given contact information, selects among three strategies (compliant MP response only; release and bound the contacted cable; release plus workspace boundary repulsion). The decision is driven by a robust release-feasibility test that checks, before any tension is reduced, whether the contacted cable can be kept below a prescribed bound at the current pose under bounded tension and length-tracking uncertainties.
- **Elasto-plastic compliance for workflow continuity.** The proposed MP-level compliance law continuously updates

a plastic equilibrium and includes an explicit restoration term. It links reflexive safety reaction, sustained cooperative guidance, and return to nominal motion using only cable-tension–derived wrench estimates. It extends prior power-based Elasto-Plastic Robot Compliance (EPRC) [35] with a continuous, tunable update and automatic recovery..

- **Contacted-cable tension regulation below a bound.** Rather than driving the contacted cable to slackness, the tension is regulated below a prescribed admissible bound $\bar{\tau}_c$ via a scheduled Quadratic Programming (QP) upper bound and a reference override on the corresponding servomotor, preserving wrench feasibility and making the cable usable as a handle for collaboration.
- **Systems-level validation across three scales.** An interaction-flow experiment validates unified transition handling across interaction modes; repeated-task trials quantify compliance transparency, force consistency, including a dedicated dynamic comparison of the model-based wrench estimate with a handle-mounted F/T sensor, and non-amplifying power exchange at the MP; and user trials with $N = 23$ novices assess usability and collaborative accessibility.

The remainder of this paper is organised around the supervisory decision. Section II scopes the interaction problem, summarises the framework, and describes the [PlatformName] platform. Section III gives only the robot modelling and nominal control details needed for the later logic. Section IV describes the perception layer (detection, distinction, identification) mostly as background, since these modules are adopted from prior work. Section V develops the novel contributions in the following order: the feasibility-aware supervisor, the elasto-plastic compliance controller, and the bounded contacted-cable regulation law. Section VI validates the framework at three complementary levels: an integrated interaction scenario demonstrating supervisor switching across cable collision, cable collaboration, and MP manipulation; repeated physical-guidance trials quantifying interaction wrench reconstruction and MP compliance; and a multi-participant user study evaluating usability and repeatability. Section VII concludes.

II. PROBLEM FORMULATION AND FRAMEWORK OVERVIEW

This section formulates the interaction problem addressed in this paper and outlines, at a high level, how the proposed framework decomposes it. It also describes the [PlatformName] platform used for evaluation.

A. Interaction Problem and Scope

We focus on single-class interaction events: contact with one cable, or with the MP, but not both concurrently. Within the MP class, the compliance law integrates the resultant wrench irrespective of the number of contact regions; within the cable class, contacts are restricted to a single cable at a time. The framework relies solely on cable-tension sensing, without direct force and moment measurement. In contrast to [25], where

TABLE I: Contribution summary by framework block.

Framework block	Status	Content in this paper
Contact perception	Reused	Contact detection, contact distinction, and contacted-cable identification reused as supervisor inputs [25].
Wrench estimation	Adapted	Standard model-based wrench reconstruction with a smoothing stage for stable tension-only compliance.
Feasibility-aware supervisor	New	Robust release-feasibility logic with strategy selection under workspace release-permission constraints.
Elasto-plastic compliance	New	MP-level compliance with adaptive plastic equilibrium and smooth recovery for pHRI.
Bounded contacted-cable regulation	New	Cable tension reduction and motor-side override to keep the contacted cable below an admissible bound.

cables are primarily treated as hazards during carrying tasks, this paper deliberately treats a cable as a potential *controlled interaction channel*, so the same supervisor and compliance law must handle all four cells of the (element) $\times$ (intent) taxonomy.

(i) *Human–cable contacts*:

- *Safety-critical human–cable collisions*. The operator unintentionally hits, brushes, or leans on a cable. The event must be detected, localised to the involved cable, and managed by limiting the contacted-cable tension while preserving the feasibility of the required MP wrench under the cable-tension constraints.
- *Collaborative human–cable contacts*. The operator intentionally grasps a cable to guide the MP. The same contacted-cable signal pathway is reused, but the management target changes from damage mitigation to sustained cooperative guidance.



Fig. 2: Human–cable physical interaction: the operator pulls the cable to bring the MP closer.

(ii) *Human–MP contacts*:

- *Safety-critical human–MP collisions*. The operator unintentionally contacts the MP frame. The event must be detected and managed by MP-level compliance while keeping the cable tensions within bounds.
- *Collaborative human–MP contacts*. The operator deliberately maintains contact with the MP to guide or co-manipulate. Here, no cable is released; the same compliance law produces a smoothly updated equilibrium that supports sustained guidance.

B. Framework Roadmap

Fig. 4 summarises the proposed framework as a decision-driven pipeline. This subsection is deliberately kept as a lightweight roadmap; each block is fully specified later and is not previewed in detail here.

The architecture has five layers: a **Perception and Estimation** layer (green) that, using only cable-tension measurements τ_m , detects contact, infers contact type and contacted-cable index c , and produces a filtered interaction-wrench

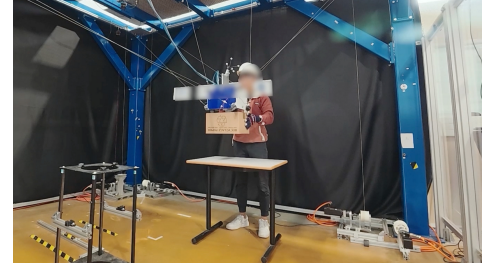


Fig. 3: Human–MP physical interaction: the operator adjusts the position of the MP through direct physical contact.

estimate $\hat{\mathbf{w}}_{\text{int}}$ (Section IV); an **Interaction Strategy Supervisor** (yellow) that, given these signals and a robust release-feasibility test, selects one of three strategies (Subsection V-A); an **Elasto-plastic Compliance Controller** (purple) that reshapes the nominal reference into a compliant command driven by $\hat{\mathbf{w}}_{\text{int}}$ and, when active, a virtual-wall term (Subsection V-B); a **Tension Distribution** block (red) that solves a QP under strategy-dependent bounds (Subsection V-C); and a **Joint-Space Controller** (blue) that synthesises motor torques with a cable- c reference override when the supervisor activates release (Subsection III-B).

The supervisor’s decision logic can be stated compactly as:

- **Strategy A (compliance only)**. Used for human–MP contacts, and for human–cable contacts where the release-feasibility test fails. Nominal tension bounds are preserved, no cable-release override is commanded, and the MP becomes compliant via the elasto-plastic controller. In the human–cable collision case, the resulting displacement is therefore an MP-level separation response: the MP and the contacted cable move away from the operator/contact region until the contact force decays, rather than trying to keep the MP fixed.
- **Strategy B (release + compliance)**. For human–cable contacts where release is robustly feasible and the MP is away from the workspace boundary. The QP upper bound of cable c is scheduled down to $\bar{\tau}_c$ and servomotor c switches to a tension-regulation override.
- **Strategy C (release + virtual wall + compliance)**. Activated when, under Strategy B, the MP approaches the workspace boundary. A virtual-wall force $\mathbf{f}_{\text{wall}}$ is injected into the compliance admittance to steer the commanded reference back toward the interior of the feasible set. Thus, boundary enforcement is handled at the reference-generation level, before the command is passed to the tension-distribution and joint-space control layers.

The backbone control architecture, consisting of the nominal



Preprint submitted to IEEE Transactions on Robotics. Received: May 21, 2026 06:26:53 Pacific Time

where $\mathbf{x} \in \mathbb{R}^3$ denotes the Cartesian position of the MP expressed in the base frame, $\boldsymbol{\omega} \in \mathbb{R}^3$ is the MP angular velocity expressed in the base frame, and l_i denotes the length of cable i between its attachment point and the corresponding pulley exit point. These scalar lengths are collected in $\mathbf{l} = [l_1, \dots, l_k]^\top$. The matrix $\mathbf{W} \in \mathbb{R}^{6 \times k}$ is the wrench matrix.

3) *Dynamic Modelling*: The rigid-body dynamic model provides the relation between cable tensions, MP motion, gravity, and external interaction wrenches. It is recalled here to define the nominal feed-forward wrench used by the tension-space controller. Following [40], the dynamic model of a CDPR is

$$\mathbf{W}\boldsymbol{\tau} - \mathbb{I}_p \dot{\mathbf{v}} - \mathbf{C}\mathbf{v} + \mathbf{w}_{\text{int}} + \mathbf{w}_g = \mathbf{0}, \quad (2)$$

where $\boldsymbol{\tau} \in \mathbb{R}^k$ is the cable tension vector, $\mathbb{I}_p$ is the spatial inertia of the platform, and $\mathbf{C}$ is the matrix of centrifugal and Coriolis wrenches. The term $\mathbf{w}_{\text{int}} \in \mathbb{R}^6$ represents the interaction wrench, including collisions and physical contacts, and $\mathbf{w}_g$ is the gravitational wrench.

The spatial inertia $\mathbb{I}_p$ is defined as

$$\mathbb{I}_p = \begin{bmatrix} m\mathbf{I}_{3 \times 3} & -[\mathbf{MS}_p]_\times \\ [\mathbf{MS}_p]_\times & \mathbf{I}_p \end{bmatrix}, \quad (3)$$

where m is the platform mass, $\mathbf{I}_{3 \times 3}$ denotes the 3×3 identity matrix in $\mathbb{R}^3$, $\mathbf{MS}_p$ is the first momentum of the MP with respect to the MP frame, and $\mathbf{I}_p$ is the inertia tensor of the platform. The skew-symmetric matrix associated with $\mathbf{MS}_p$ is denoted by $[\mathbf{MS}_p]_\times$. The inertia tensor $\mathbf{I}_p$ is obtained from the inertia tensor $\mathbf{I}_g$ in the MP frame using the Huygens–Steiner theorem:

$$\mathbf{I}_p = \mathbf{R}_p \mathbf{I}_g \mathbf{R}_p^T - \frac{[\mathbf{MS}_p]_\times [\mathbf{MS}_p]_\times}{m}. \quad (4)$$

The gravity wrench $\mathbf{w}_g$ due to gravity vector $\mathbf{g} = [0, 0, -g]^\top$ is

$$\mathbf{w}_g = \begin{bmatrix} m\mathbf{I}_{3 \times 3} \\ [\mathbf{MS}_p]_\times \end{bmatrix} \mathbf{g}, \quad (5)$$

where $g = 9.81 \text{ m s}^{-2}$. In (2), the matrix $\mathbf{C}$ yields centrifugal and Coriolis wrenches:

$$\mathbf{C}\mathbf{v} = \begin{bmatrix} [\boldsymbol{\omega}]_\times [\boldsymbol{\omega}]_\times \mathbf{MS}_p \\ [\boldsymbol{\omega}]_\times \mathbf{I}_p \boldsymbol{\omega} \end{bmatrix}. \quad (6)$$

In nominal operation, the interaction wrench is set to zero in (2). This gives the model-based feed-forward wrench used by the tension-space controller below.

B. Nominal Control Architecture

The nominal controller has two coupled parts. The tension-space controller computes desired cable tensions $\boldsymbol{\tau}_d$ by solving a QP based on a model-based feed-forward wrench. The joint-space controller then computes the desired motor joint angles $\mathbf{q}_d$ through the IGEM and applies a PID torque correction. These two parts define the nominal baseline on which the interaction supervisor acts in Section V.

1) *Tension-Space Controller*: To generate a feasible nominal tension distribution, a model-based feed-forward wrench

$\mathbf{w}_{\text{req}}$ is derived from (2) by setting $\mathbf{w}_{\text{int}} = \mathbf{0}$:

$$\mathbf{w}_{\text{req}} = \mathbb{I}_p \dot{\mathbf{v}}_d + \mathbf{C}\mathbf{v}_d - \mathbf{w}_g, \quad (7)$$

where $\mathbf{v}_d$ and $\dot{\mathbf{v}}_d$ are the desired MP twist and twist acceleration, respectively. In this nominal feed-forward term, no direct contact-wrench compensation is applied; contact effects are instead handled by the compliance and interaction-management layers.

Let $\mathbf{W}_d \triangleq \mathbf{W}(\mathbf{p}_d) \in \mathbb{R}^{6 \times k}$ denote the wrench matrix calculated at the desired MP pose $\mathbf{p}_d$. With the sign convention adopted in (1), the desired cable tensions are parameterised as

$$\boldsymbol{\tau}_d(\boldsymbol{\lambda}) = -\mathbf{W}_d^+ \mathbf{w}_{\text{req}} + \mathbf{N}_d \boldsymbol{\lambda}, \quad (8)$$

where $(\cdot)^+$ denotes the Moore–Penrose pseudo-inverse, $\mathbf{N}_d \in \mathbb{R}^{k \times (k-6)}$ is a basis for the null space of $\mathbf{W}_d$, and $\boldsymbol{\lambda} \in \mathbb{R}^{k-6}$ is the vector of internal tension parameters. By construction, $\mathbf{W}_d \mathbf{N}_d = \mathbf{0}_{6 \times (k-6)}$, so the term $\mathbf{N}_d \boldsymbol{\lambda}$ modifies only the internal cable tensions and does not change the resultant wrench at the MP.

The internal tension vector is selected by solving the following QP [36]:

$$\begin{aligned} \boldsymbol{\lambda}^* = \arg \min_{\boldsymbol{\lambda} \in \mathbb{R}^{k-6}} \quad & \|\boldsymbol{\tau}_d(\boldsymbol{\lambda})\|^2, \\ \text{subject to} \quad & \boldsymbol{\tau}_{\min} \leq \boldsymbol{\tau}_d(\boldsymbol{\lambda}) \leq \boldsymbol{\tau}_{\max}. \end{aligned} \quad (9)$$

The desired cable tension vector is then

$$\boldsymbol{\tau}_d = \boldsymbol{\tau}_d(\boldsymbol{\lambda}^*). \quad (10)$$

For the fully constrained CDPR considered in this work, $k = 8$ cables actuate a rigid platform in $SE(3)$. Therefore, $\mathbf{W}_d \in \mathbb{R}^{6 \times 8}$, $\dim \text{Null}(\mathbf{W}_d) = 8 - 6 = 2$, and $\boldsymbol{\lambda} \in \mathbb{R}^2$. The lower bound $\boldsymbol{\tau}_{\min}$ ensures cable tautness by maintaining a minimum tension of 1 N per cable, while $\boldsymbol{\tau}_{\max}$ sets the admissible upper tension limit to 100 N per cable.

2) *Joint-Space Controller*: The desired motor joint angles $\mathbf{q}_d$ are computed from the desired MP pose $\mathbf{p}_d$ and the desired cable tensions $\boldsymbol{\tau}_d$ using the IGEM, thereby accounting for the geometric and elasto-static constraints of the system. A PID corrector with gains K_p , K_i , and K_d forces the tracking error $\mathbf{e}_q = \mathbf{q} - \mathbf{q}_d$ to converge to zero with

$$\ddot{\mathbf{e}}_q + K_d \dot{\mathbf{e}}_q + K_p \mathbf{e}_q + K_i \int \mathbf{e}_q dt = \mathbf{0}. \quad (11)$$

The motor joint acceleration command is defined as

$$\ddot{\mathbf{q}} = \ddot{\mathbf{q}}_d - K_d (\dot{\mathbf{q}} - \dot{\mathbf{q}}_d) - K_p (\mathbf{q} - \mathbf{q}_d) - K_i \int (\mathbf{q} - \mathbf{q}_d) dt. \quad (12)$$

The corrective torque generated by the joint-space controller is

$$\boldsymbol{\Gamma}_c = \mathbf{I}_m \ddot{\mathbf{q}} + \boldsymbol{\Gamma}_f, \quad (13)$$

where $\mathbf{I}_m$ is the motor–gearbox inertia matrix, referred to the motor side, and $\boldsymbol{\Gamma}_f$ compensates for dominant Coulomb and viscous friction. The friction-compensation torque vector is defined as

$$\boldsymbol{\Gamma}_f \triangleq \mathbf{f}_s \odot \text{sign}(\dot{\mathbf{q}}_d) + \mathbf{f}_v \odot \dot{\mathbf{q}}_d, \quad (14)$$

where $\odot$ denotes element-wise multiplication. The vectors $\mathbf{f}_s$ and $\mathbf{f}_v$ are the identified Coulomb and viscous friction

coefficients, respectively.

The resulting motor–gearbox torque command combines the feed-forward torque associated with the desired cable tensions and the corrective joint-space torque:

$$\Gamma_{\text{mg}} = \text{sat}(r_d \tau_d + \Gamma_c), \quad (15)$$

where $\text{sat}(\cdot)$ denotes the motor-torque saturation function and r_d is the winch radius.

The quantities introduced in this nominal architecture are reused later by the interaction framework through two interfaces. First, the dynamic balance in (2) is inverted in Section IV-B to reconstruct the external interaction wrench from measured cable tensions. Second, the tension-distribution equations (8)–(9) is reused in Section V with strategy-dependent tension bounds, in particular when the contacted cable is regulated under Strategies B and C.

IV. CONTACT PERCEPTION

The supervisor of Section V contains four perception outputs: a contact flag, a contact-type label (human–cable vs. human–MP), the contacted-cable index c , and a filtered external-wrench estimate $\tilde{\mathbf{w}}_{\text{int}}$. Accordingly, this section presents the contact-detection, contact-distinction, and cable-identification modules, and then introduces the smoothing stage used in the external-wrench estimator.

A. Contact Detection

We use the tension-error-based detection method of [27]. Let τ_{mf} denote the filtered measured cable tensions and τ_d the desired tensions computed by the nominal tension-distribution controller. The tension-tracking error is defined as

$$\delta\tau = \tau_{\text{mf}} - \tau_d. \quad (16)$$

A contact is declared when at least one component of $\delta\tau$ exceeds a threshold $T_h = 4\text{ N}$:

$$\exists i \in \{1, \dots, k\} \quad \text{such that} \quad |\delta\tau_i| > T_h. \quad (17)$$

The threshold is chosen as $T_h = \delta_{\text{max}} + \Delta_s$, with $\delta_{\text{max}} \approx 2\text{ N}$ the maximum tension error observed under nominal conditions, including sensor noise and residual model mismatch, and $\Delta_s = 2\text{ N}$ a safety margin. This conservative choice trades a short detection delay for reliable rejection of false positives, consistent with [18]. Once condition (17) is satisfied, the framework switches from nominal operation to the perception pipeline, where the contact type and, when relevant, the contacted cable are estimated.

B. External Wrench Estimator

The model-based reconstruction below is standard; the main novelty in this subsection is the smoothing stage $\mathcal{F}_{\text{DA}}$, introduced to mitigate the adverse wrench–compliance coupling described later. Based on the dynamic model in (2), the raw interaction wrench acting on the MP is reconstructed as $\tilde{\mathbf{w}}_{\text{int}}$ by evaluating the model at the desired state:

$$\tilde{\mathbf{w}}_{\text{int}} = \mathbb{I}_p(\mathbf{p}_d) \dot{\mathbf{v}}_d + \mathbf{C}(\mathbf{p}_d, \mathbf{v}_d) \mathbf{v}_d - \mathbf{w}_g(\mathbf{p}_d) - \mathbf{W}(\mathbf{p}_d) \tau_{\text{mf}}. \quad (18)$$

The raw interaction wrench is defined as

$$\tilde{\mathbf{w}}_{\text{int}} = \begin{bmatrix} \tilde{\mathbf{f}}_{\text{int}} \\ \tilde{\mathbf{m}}_{\text{int}} \end{bmatrix} = \begin{bmatrix} \tilde{f}_{\text{int}}^x & \tilde{f}_{\text{int}}^y & \tilde{f}_{\text{int}}^z & \tilde{m}_{\text{int}}^x & \tilde{m}_{\text{int}}^y & \tilde{m}_{\text{int}}^z \end{bmatrix}^\top, \quad (19)$$

where $\tilde{f}_{\text{int}}^i$ and $\tilde{m}_{\text{int}}^i$ denote the raw interaction force and moment components along Cartesian axis i , respectively. For the human–cable interactions considered here, the dominant contribution is the force component $\tilde{f}_{\text{int}}^x$. Although cable tensions may also induce a moment about the MP geometric centre, this contribution is typically small compared with $\tilde{f}_{\text{int}}^x$ and is therefore not used by the compliance law.

A key difficulty in this setting is that the interaction wrench is reconstructed from cable-tension measurements and then reused in the compliant control law. This estimation–action loop makes the closed-loop behaviour sensitive to measurement noise, model mismatch, and tension redistribution. As a result, a standard admittance law driven directly by the raw tension-derived wrench is not sufficient here. The proposed smoothing stage is therefore not merely a refinement, but a necessary element for achieving stable and interpretable tension-only compliance.

A key feature of (18) is that it does not rely on absolute MP pose feedback in the control loop. The velocity $\mathbf{v}_d$ and acceleration $\dot{\mathbf{v}}_d$ are obtained from the desired/compliant trajectory (cf. Section II-B), not from measured MP motion. Consequently, modelling errors, tension-sensor noise, and discrepancies between desired and actual motion appear directly in the raw interaction wrench $\tilde{\mathbf{w}}_{\text{int}}$ (and in particular in $\tilde{\mathbf{f}}_{\text{int}}$), without being compensated by an independent absolute pose measurement.

In practice, the raw estimate $\tilde{\mathbf{f}}_{\text{int}}$ exhibits a small static offset (typically below 2 N) due to auxiliary elements such as communication cables, together with measurement noise and small oscillations. Beyond these issues, a more critical difficulty arises from the coupling between wrench estimation and compliance control: the nominal compliance law reshapes the trajectory based on the interaction-wrench signal, which modifies $\mathbf{W}_d$ and feeds back into (18). Moreover, the resulting compliant motion also perturbs the measured tensions τ_{mf} (via tension redistribution and the joint-space controller), such that the wrench estimate is driven by signals that are themselves affected by the compliance action. In the worst case, oscillations in τ_{mf} induce oscillations in $\tilde{\mathbf{f}}_{\text{int}}$, which in turn cause rapid variations of the compliant trajectory and of $\mathbf{W}_d$, forming an adverse feedback mechanism that can destabilise the closed loop.

This risk partly explains why pHRI controllers for CDPRs based solely on cable-tension-derived wrench estimation are seldom reported: many approaches rely on additional force sensing or high-accuracy absolute pose feedback to break such couplings. In the present framework, stable and intuitive behaviour requires the interaction-wrench signal used for compliance control, denoted $\tilde{\mathbf{w}}_{\text{int}}$, to be sufficiently smooth and free of small-offset chatter. Accordingly, the raw wrench $\tilde{\mathbf{w}}_{\text{int}}$ is pre-processed using a lightweight deadzone plus Adaptive Exponential Moving Average (AEMA) filter [41]:

$$\hat{\mathbf{w}}_{\text{int}} = \mathcal{F}_{\text{DA}}(\tilde{\mathbf{w}}_{\text{int}}), \quad (20)$$

where $\mathcal{F}_{\text{DA}}(\cdot)$ denotes the proposed deadzone–AEMA filter. It suppresses small static offsets and high-frequency chatter while preserving responsiveness to genuine human inputs. Figure 5 illustrates the resulting smoothing effect, and the compliance controller uses $\hat{\mathbf{w}}_{\text{int}}$ as its driving interaction input.

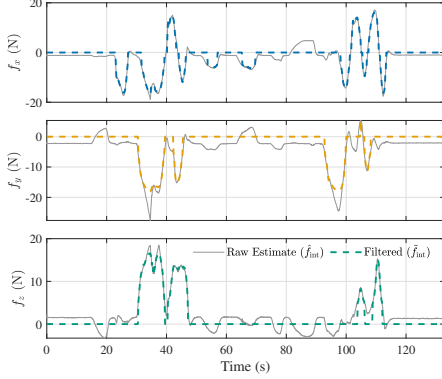


Fig. 5: Comparison between the raw interaction force estimate $\tilde{\mathbf{f}}_{\text{int}}$ (from $\tilde{\mathbf{w}}_{\text{int}}$) and the pre-processed (filtered) interaction force $\hat{\mathbf{f}}_{\text{int}}$ (from $\hat{\mathbf{w}}_{\text{int}}$) using the proposed deadzone and AEMA filter.

C. Contact Distinction

This paper reuses the contact-distinction pipeline of [25], whose output label (human–cable vs. human–MP) is a required input for the supervisor. It combines the model-based wrench estimate of Section IV-B with Spectral Norm Derivative (SND) features and a deterministic scoring classifier, exploiting the observation [23], [24], [42] that deliberate human forces and collision impulses have distinct spectral signatures. The SND feature definition and the resulting classification principle are summarised here, and the calibrated thresholds and scoring weights are inherited from [25].

Let $\tilde{f}_{\text{int}}^j(t)$ and $\tilde{m}_{\text{int}}^j(t)$ denote the j -th Cartesian components, with $j \in \{x, y, z\}$, of the raw interaction force $\tilde{\mathbf{f}}_{\text{int}}$ and raw interaction moment $\tilde{\mathbf{m}}_{\text{int}}$ in (19), respectively. Over a sliding window of N samples, with sampling time T_s , a Fast Fourier Transform (FFT) is computed using a half-Hann weighting and restricted to a frequency band $[\omega_{\min}, \omega_{\max}] = [3, 50]$ Hz. This band captures contact-induced dynamics while rejecting low-frequency MP motion and high-frequency noise. The corresponding SND indices are defined as

$$I_{\mathbf{f}}^j(t) = \frac{d}{dt} \left\| \tilde{\mathbf{F}}_{[\omega_{\min}, \omega_{\max}]}^j(t) \right\|_1, \quad (21)$$

$$I_{\mathbf{m}}^j(t) = \frac{d}{dt} \left\| \tilde{\mathbf{M}}_{[\omega_{\min}, \omega_{\max}]}^j(t) \right\|_1, \quad (22)$$

where $\tilde{\mathbf{F}}_{[\omega_{\min}, \omega_{\max}]}^j$ and $\tilde{\mathbf{M}}_{[\omega_{\min}, \omega_{\max}]}^j$ collect the band-limited magnitude-spectrum samples for the force and moment channels. The ℓ_1 -aggregation improves robustness to narrow-band peaks, such as structural resonances, while the time derivative captures rapid changes in spectral energy that are characteristic of impacts.

In [25], the indices in (21)–(22) are compared with calibrated force and moment thresholds, and the resulting Boolean indicators are combined into weighted scenario scores. The label with the highest score is selected as the contact type. In brief, human–cable contact typically yields comparatively small SND indices due to cable damping, whereas human–MP contact generates stronger transient signatures in the force and moment channels. This rule-based criterion reduces ambiguity when several indicators are activated simultaneously and provides the contact-type label used by the supervisor.

D. Cable Collision Identification

When the contact-distinction module reports a human–cable event, the specific contacted cable must be identified so that Strategies B/C of Section V-A can act on the correct QP bound and servomotor override. A naive $\arg \max_i |\delta \tau_i|$ heuristic is unreliable because a local interaction on one cable redistributes tensions through the closed-loop wrench balance, so non-contacted cables may exhibit larger deviations than the contacted one. The identifier below sidesteps this ambiguity.

Following [36], cable collision identification is formulated by exploiting a geometric consistency property: under the assumption that (i) a single cable c undergoes a shape change during the collision and (ii) the MP remains close to its desired pose during the event, only the direction of the collided cable deviates significantly from its desired direction. Let $\mathbf{u}_{id} \in \mathbb{R}^3$ denote the desired unit direction vector of cable i (computed from the IGEM using the desired MP pose), and let $\delta \tau = [\delta \tau_1, \dots, \delta \tau_k]^\top$ be the tension variation vector over a short interval around the detected contact. Since the MP remains approximately stable, the net change of translational cable forces on the MP is assumed to be negligible, yielding

$$\mathbf{U}_r \delta \tau \approx \mathbf{0}_3, \quad (23)$$

where $\mathbf{U}_r = [\mathbf{u}_{1r}, \dots, \mathbf{u}_{kr}]$ collects the (unknown) actual cable directions. Using the single-collision assumption, $\mathbf{u}_{ir} \approx \mathbf{u}_{id}$ for $i \neq c$, while $\mathbf{u}_{cr}$ may differ. Equation (23) can then be rearranged as

$$\sum_{i=1, i \neq c}^k \mathbf{u}_{id} \delta \tau_i \approx -\mathbf{u}_{cr} \delta \tau_c. \quad (24)$$

Taking norms and using $\|\mathbf{u}_{cr}\|_2 = 1$ gives the key identification relation

$$\left\| \sum_{i=1, i \neq c}^k \mathbf{u}_{id} \delta \tau_i \right\|_2 \approx |\delta \tau_c|. \quad (25)$$

Accordingly, [36] defines a geometric-consistency metric

$$g(r) = \left\| \sum_{i=1, i \neq r}^k \mathbf{u}_{id} \delta \tau_i \right\|_2 - |\delta \tau_r|, \quad r \in \{1, \dots, k\}, \quad (26)$$

which ideally satisfies $g(c) \approx 0$ for the collided cable index c . In practice, identification is refined through a composite criterion: likely cables are first selected from the largest tension deviations $|\delta \tau_i|$ (to ensure reactivity and robustness to noise), and then disambiguated by selecting the cable with the smallest

$g(r)$ values [36]. This compound strategy substantially reduces mis-identification compared to a single $|\delta\tau_i|$ -based heuristic and provides a reliable collided-cable index for the subsequent tension-release module.

V. FEASIBILITY-AWARE INTERACTION MANAGEMENT

This section develops the three novel components of the paper following the sequence of the supervisor's decision flow: first, the supervisor and its robust release-feasibility test (Subsection V-A); then the elasto-plastic compliance law that produces the MP response common to all strategies (Subsection V-B); and finally the contacted-cable tension regulation invoked by Strategies B/C (Subsection V-C). The central design principle is that *releasing a contacted cable is a privilege, not a default reaction*. The supervisor grants this privilege only where the release can be robustly enforced; everywhere else, the operator still experiences compliant behaviour, but through the MP only.

To make this privilege explicit, we use the restrained feasible workspace with safety margin (RFWSM) as a **permission boundary**: a precomputed region of poses in which the contacted-cable tension can be kept below or around a prescribed bound $\bar{\tau}_c$ under bounded tension and length-tracking errors. Inside the RFWSM, the supervisor enables Strategies B/C (release + compliance, possibly with a virtual wall near the boundary); outside, it must fall back to Strategy A (compliance only, nominal tension bounds). The RFWSM is thus not a workspace descriptor in the classical sense; it is the safe-release permission set on which the supervisor relies.

A. Interaction Strategy Supervisor

1) *RFWSM as a release-permission set*: The RFWSM is the set of MP poses where the robust constraint $\tau_c \leq \bar{\tau}_c$ can be satisfied under bounded tension and length-tracking errors while still meeting static wrench equilibrium and the global tension bounds. Following standard CDPR workspace-determination procedures [1], [43], it is obtained offline by discretising the MP pose domain and applying, at each pose, the robust wrench-feasibility test defined in (29); each pose is thereby labelled as release-feasible or not, with no classical scalar capacity margin involved. Inside the RFWSM, the supervisor grants Strategies B/C; outside, it reverts to Strategy A. Unlike prior cable-release formulations that drive the contacted cable to slackness [27], the present framework requires the contacted-cable tension to remain below or around $\bar{\tau}_c$ while preserving unilateral actuation. In implementation, one RFWSM is precomputed offline for each possible contacted cable $c \in \{1, \dots, k\}$, and the supervisor uses the RFWSM associated with the detected cable.

Let $\boldsymbol{\tau} \in \mathbb{R}^k$ denote the cable-tension vector ($k = 8$), with τ_c the tension of cable c .

2) *Uncertainty bound*: We model bounded tension-tracking and cable-length errors by $\mathbf{e}_\tau$ and $\mathbf{e}_l$, and aggregate them into the conservative tension uncertainty bound $\mathbf{u}_\tau \in \mathbb{R}^k$:

$$\mathbf{u}_\tau = |\mathbf{e}_\tau| + \mathbf{K}_s |\mathbf{e}_l|, \quad (27)$$

where $\mathbf{K}_s = \text{diag}(ES/l_i)$ is the axial-stiffness matrix associated with the current cable lengths $\mathbf{l}$. The product ES characterises the cable elasticity, while ES/l_i gives the corresponding axial-stiffness coefficient for cable i ; ES is identified experimentally in the setup, following the procedure used in [12], [14]. The vector $\mathbf{l}$ contains the cable lengths from the motor gearboxes to the MP anchor points, reconstructed from motor-encoder measurements using the winch/drum and pulley geometry.

3) *Worst-case contacted-cable perturbation*: In [44], [45], wrench feasibility under bounded tension- and length-induced errors is assessed by evaluating all 2^k sign combinations of cable-error realisations. While conservative, this exhaustive vertex enumeration is not suitable for online contact management. We therefore adopt a low-dimensional worst-case analysis that directly computes the maximum admissible upward tension perturbation on the contacted cable c , denoted $\nu_c^{\max}$, under equilibrium-preserving perturbations (i.e., perturbations restricted to the null space of the wrench matrix $\mathbf{W}$).

Let $\boldsymbol{\nu} \in \mathbb{R}^k$ denote an admissible tension perturbation and $\mathbf{e}_c \in \mathbb{R}^k$ the c -th canonical basis vector. The maximum physically admissible increase in the contacted-cable tension is obtained from

$$\begin{aligned} \nu_c^{\max} &= \max_{\boldsymbol{\lambda}} \mathbf{e}_c^T \mathbf{N} \boldsymbol{\lambda} \\ \text{subject to } \boldsymbol{\nu} &= \mathbf{N} \boldsymbol{\lambda}, \\ &-\mathbf{u}_\tau \leq \boldsymbol{\nu} \leq \mathbf{u}_\tau, \end{aligned} \quad (28)$$

where $\mathbf{N}$ is a basis matrix of $\text{Null}(\mathbf{W})$, i.e., $\mathbf{W}\mathbf{N} = \mathbf{0}$.

4) *Online RFWSM feasibility test and release-feasible flag*: For a given MP pose $\mathbf{p}$ and a candidate contacted cable c , the RFWSM feasibility test checks whether there exists a nominal tension vector $\boldsymbol{\tau}$ that satisfies static wrench equilibrium, the global cable-tension bounds, and the robust upper bound on the contacted cable:

$$\begin{aligned} \text{find } \boldsymbol{\tau} \\ \text{subject to } \mathbf{W}(\mathbf{p})\boldsymbol{\tau} &= \mathbf{w}_{\text{req}}(\mathbf{p}), \\ \boldsymbol{\tau}_{\min} &\leq \boldsymbol{\tau} \leq \boldsymbol{\tau}_{\max}, \\ \tau_c + \nu_c^{\max}(\mathbf{p}) &\leq \bar{\tau}_c. \end{aligned} \quad (29)$$

The last inequality is the robust contacted-cable constraint: the nominal tension τ_c must remain below the prescribed bound $\bar{\tau}_c$ with a margin $\nu_c^{\max}$ reserved for worst-case tension and length-tracking errors. Equivalently, the nominal contacted-cable tension is constrained by the tightened upper bound

$$\tau_c \leq \bar{\tau}_c - \nu_c^{\max}(\mathbf{p}).$$

The release-feasible flag is then defined directly from the existence of a solution to (29):

$$s_{\text{rel}}(\mathbf{p}, c) = \begin{cases} 1, & \text{if (29) is feasible,} \\ 0, & \text{otherwise.} \end{cases} \quad (30)$$

Let $\Omega_h = \{\mathbf{p}_j\}_{j=1}^{N_h}$ denote the discretised pose grid. The discretised RFWSM associated with cable c is the subset

$$\text{RFWSM}_c = \{\mathbf{p} \in \Omega_h : s_{\text{rel}}(\mathbf{p}, c) = 1\}, \quad (31)$$

i.e., the poses for which a tension vector exists that satisfies

static equilibrium, the cable-tension bounds, and the tightened contacted-cable constraint $\tau_c + \nu_c^{\max}(\mathbf{p}) \leq \bar{\tau}_c$. No classical scalar capacity margin is used: each pose is classified by binary feasibility, with $\nu_c^{\max}$ acting as the robustness margin on the contacted-cable tension.

The binary flag $s_{\text{rel}}(\mathbf{p}, c)$ is stored in the precomputed lookup table of cable c . Online, once the contacted cable has been identified, the supervisor queries the corresponding table at the current desired or compliant pose. If $s_{\text{rel}} = 1$, Strategies B/C are enabled; if $s_{\text{rel}} = 0$, cable release is disabled and the supervisor reverts to Strategy A.

5) *Experimental workspace illustration (RFW vs. RFWSM)*: In the experiments on the [PlatformName] platform, cable elasticity is characterised by the identified product $ES = 17732$ N, a bounded tension-tracking error $\mathbf{e}_\tau = 2$ N (component-wise), and a bounded cable-length tracking error $\mathbf{e}_l = 1$ mm. During contact management, the contacted cable is constrained to remain below $\bar{\tau}_c = 25$ N. This value was chosen as a compromise between safety and perception: it remains low enough to limit the contact force applied to the human, while preserving sufficient cable tension for reliable contact perception and interaction management. The corresponding restrained feasible workspace (RFW) and its robust counterpart RFWSM are precomputed for all cables; Fig. 6 shows the representative case $c = 5$.

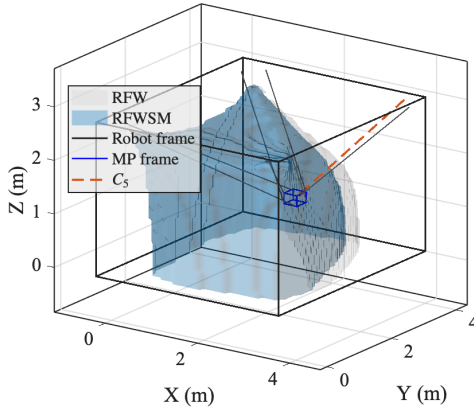


Fig. 6: RFW (nominal) and RFWSM (robust) under the contacted-cable constraint $\tau_c \leq \bar{\tau}_c$ with $\bar{\tau}_c = 25$ N ($c = 5$). The RFWSM volume is reduced by 7.78% relative to the RFW.

The anchor-point layout on the robot frame is symmetric, so selecting $c = 5$ yields a representative workspace reduction that is qualitatively similar for the other cables under comparable contact conditions. The volume of a workspace $\mathcal{S}$ is estimated as $V(\mathcal{S}) = N_S \Delta V$, where N_S is the number of feasible voxels in Ω_h and ΔV is the voxel volume. In this setting, the RFWSM reduces the workspace volume from $V_{\text{RFW}} = 22.38$ m³ to $V_{\text{RFSM}} = 20.64$ m³, corresponding to a shrinkage of 7.78%, while exhibiting no subset violations ($\text{RFSM} \subseteq \text{RFW}$). Practically, the RFWSM excludes poses that are nominally feasible but where realised uncertainty could push τ_c above $\bar{\tau}_c$; in those poses, cable release is not permitted.

6) *Enforcing the permission boundary: virtual wall + projection (Strategy C)*: The set RFSM_c specifies the poses

where release of cable c is robustly feasible, but during compliant interaction the admittance can drift the commanded MP pose toward its boundary under persistent external forces. To keep the compliant motion within the permission set, we augment the compliance loop with a lightweight virtual wall constructed from a signed distance field (SDF) of the voxelised RFWSM.

Let $\mathbf{x} \in \mathbb{R}^3$ denote the translational MP position (orientation is set by the nominal trajectory in our experiments), and let $d(\mathbf{x})$ be the signed distance to the RFWSM boundary such that $d(\mathbf{x}) > 0$ for $\mathbf{x} \in \text{RFSM}$, $d(\mathbf{x}) = 0$ on the boundary, and $d(\mathbf{x}) < 0$ outside. The associated inward unit normal is obtained from the SDF gradient,

$$\mathbf{a}(\mathbf{x}) = \frac{\nabla d(\mathbf{x})}{\|\nabla d(\mathbf{x})\| + \varepsilon}, \quad (32)$$

where $\varepsilon > 0$ prevents division by zero in flat regions. For each precomputed RFWSM, the fields $d(\mathbf{x})$ and $\mathbf{a}(\mathbf{x})$ are computed offline on the RFWSM grid and stored as lookup tables; online, the controller queries them at the current compliant pose.

We activate a soft boundary effect within a buffer layer of thickness $d_{\text{buf}} = 0.1$ m by defining the buffer penetration depth

$$\rho(\mathbf{x}) = [d_{\text{buf}} - \max(d(\mathbf{x}), 0)]_+, \quad (33)$$

where $[\cdot]_+ = \max(\cdot, 0)$. Let $\dot{\mathbf{x}}_d$ be the compliant MP velocity and define the inward normal velocity $v_n = \mathbf{a}(\mathbf{x}_d)^\top \dot{\mathbf{x}}_d$ (so $v_n < 0$ indicates outward motion). The virtual wall force is then

$$\mathbf{f}_{\text{wall}} = K_{\text{wall}} \rho(\mathbf{x}_d) \mathbf{a}(\mathbf{x}_d) - B_{\text{wall}} \min(v_n, 0) \mathbf{a}(\mathbf{x}_d), \quad (34)$$

with gains $K_{\text{wall}} > 0$ and $B_{\text{wall}} > 0$.

To provide a hard safety guarantee against discretisation effects and finite-step integration, we additionally apply a projection step whenever the tentative compliant update $\mathbf{x}_d^*$ leaves the RFWSM (i.e., $d(\mathbf{x}_d^*) < 0$):

$$\mathbf{x}_d = \mathbf{x}_d^* - d(\mathbf{x}_d^*) \mathbf{a}(\mathbf{x}_d^*) + \varepsilon_{\text{in}} \mathbf{a}(\mathbf{x}_d^*), \quad d(\mathbf{x}_d^*) < 0, \quad (35)$$

where $\varepsilon_{\text{in}} > 0$ is a small inward offset (chosen on the order of the grid resolution) to avoid boundary chattering. Equations (34)–(35) implement a hybrid constraint mechanism: the virtual wall smooths the approach to the boundary, whereas the projection step keeps the compliant trajectory inside the feasible workspace. We define the admittance wrench used by the compliant controller as

$$\hat{\mathbf{w}}_{\text{adm}} \triangleq \begin{bmatrix} \hat{\mathbf{f}}_{\text{int}} + \mathbf{f}_{\text{wall}} \\ \mathbf{0}_3 \end{bmatrix} \in \mathbb{R}^6, \quad (36)$$

where $\hat{\mathbf{f}}_{\text{int}} \in \mathbb{R}^3$ is the filtered interaction-force estimate from (20).

B. Elasto-Plastic Compliance Controller

As a shared foundation for all three strategies, the compliance law ensures seamless workflow continuity. The same controller governs everything from the initial safety-critical reflex and sustained manual guidance to the smooth restoration of nominal movement after contact ends. Compliance here is

not a synonym for low stiffness; it is the mechanism that links safety reaction and intentional collaboration without requiring an explicit mode change visible to the user.

The proposed framework (Fig. 4) relies on the MP-level compliance module that reshapes the nominal task-space reference into an interaction-consistent trajectory. In CDPR pHRI, physical contact can occur at either the MP or along the cables. Because direct force sensing on the MP is typically unavailable, the interaction wrench must be inferred from cable tensions, leaving it susceptible to modelling and measurement uncertainties. A fixed stiff reference tracking law would therefore either oppose the operator during collaboration or amplify contact transients, whereas a compliant reference update can dissipate contact energy and make the robot response predictable to the user. The objective is twofold: (i) to comply safely under physical interaction so as to reduce the interaction wrench, and (ii) to avoid undesired snap-back to the nominal path during sustained guidance by introducing a controlled, reversible plastic deformation. The resulting behaviour is elasto-plastic: short disturbances are absorbed elastically and filtered, whereas sustained interaction produces a stable shift of the effective equilibrium, enabling collaborative guidance and smooth recovery.

1) *Admittance Core with an Updated Equilibrium:* The MP configuration is represented by the task-space pose vector

$$\mathbf{p} = \begin{bmatrix} \mathbf{x} \\ \boldsymbol{\theta} \end{bmatrix} \in \mathbb{R}^6, \quad (37)$$

where $\mathbf{x} \in \mathbb{R}^3$ denotes the Cartesian position and $\boldsymbol{\theta} \in \mathbb{R}^3$ is the vector of MP orientation angles.

To treat orientation consistently, pose updates and pose differences are expressed with the composition operator $\oplus$ and the difference operator $\ominus$ [46]. For two poses $\mathbf{p}_1 = [\mathbf{x}_1^\top, \boldsymbol{\theta}_1^\top]^\top$ and $\mathbf{p}_2 = [\mathbf{x}_2^\top, \boldsymbol{\theta}_2^\top]^\top$, we use

$$\mathbf{p}_1 \ominus \mathbf{p}_2 \triangleq \begin{bmatrix} \mathbf{x}_1 - \mathbf{x}_2 \\ \log(\mathbf{R}(\boldsymbol{\theta}_2)^\top \mathbf{R}(\boldsymbol{\theta}_1)) \end{bmatrix} \in \mathbb{R}^6, \quad (38)$$

and, for an increment $\boldsymbol{\xi} = [\boldsymbol{\xi}_x^\top, \boldsymbol{\xi}_\theta^\top]^\top \in \mathbb{R}^6$,

$$\mathbf{p} \oplus \boldsymbol{\xi} \triangleq \begin{bmatrix} \mathbf{x} + \boldsymbol{\xi}_x \\ \log(\mathbf{R}(\boldsymbol{\theta}) \exp(\widehat{\boldsymbol{\xi}}_\theta)) \end{bmatrix}, \quad (39)$$

where $\mathbf{R}(\cdot) \in SO(3)$ is the rotation matrix, and $\log(\cdot)$ and $\exp(\cdot)$ denote the standard mappings between $SO(3)$ and $\mathbb{R}^3$.

Let $\mathbf{p}_0(t)$, $\mathbf{v}_0(t)$, and $\dot{\mathbf{v}}_0(t)$ denote the nominal MP reference produced by the planner. The compliance controller generates the commanded reference $(\mathbf{p}_d, \mathbf{v}_d)$ by simulating a virtual mass–damper–spring system driven by the admittance interaction wrench $\hat{\mathbf{w}}_{\text{adm}}$. In contrast to purely elastic compliance, the plastic term is defined with respect to an updated equilibrium (cf. Fig. 7)

$$\mathbf{p}_0^*(t) \triangleq \mathbf{p}_0(t) \oplus \mathbf{p}_{\text{pl}}(t), \quad (40)$$

where $\mathbf{p}_{\text{pl}} \in \mathbb{R}^6$ is the plastic deformation state. Define the MP pose and twist errors with respect to the updated equilibrium as

$$\boldsymbol{\xi}_p \triangleq \mathbf{p}_d \ominus \mathbf{p}_0^*, \quad \boldsymbol{\xi}_v \triangleq \mathbf{v}_d - \mathbf{v}_0, \quad (41)$$

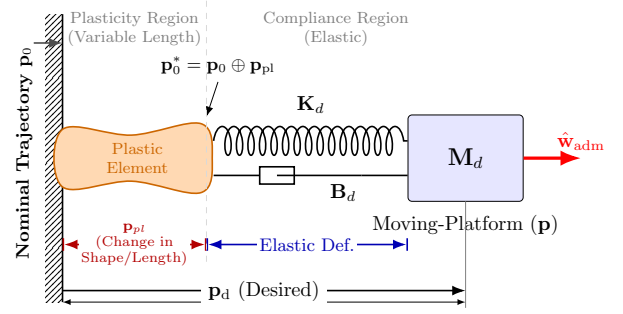


Fig. 7: Mechanical analogue of the elasto-plastic compliance concept (1D along an interaction direction). The nominal reference $\mathbf{p}_0$ acts as a fixed anchor, while the plastic state $\mathbf{p}_{\text{pl}}$ updates the equilibrium $\mathbf{p}_0^* = \mathbf{p}_0 \oplus \mathbf{p}_{\text{pl}}$. The MP pose $\mathbf{p}$ is connected to $\mathbf{p}_0^*$ through a virtual mass–damper–spring admittance $(\mathbf{M}, \mathbf{B}, \mathbf{K})$ driven by $\hat{\mathbf{w}}_{\text{adm}}$.

where $\boldsymbol{\xi}_p \in \mathbb{R}^6$ and $\boldsymbol{\xi}_v \in \mathbb{R}^6$.

The admittance dynamics is written by introducing the compliance acceleration

$$\mathbf{a}_c = \mathbf{M}^+(\hat{\mathbf{w}}_{\text{adm}} - \mathbf{B} \boldsymbol{\xi}_v - \mathbf{K} \boldsymbol{\xi}_p), \quad (42)$$

so that

$$\dot{\mathbf{v}}_d = \dot{\mathbf{v}}_0 + \mathbf{a}_c, \quad (43)$$

where $\mathbf{M}$, $\mathbf{B}$, and $\mathbf{K}$ are the active virtual inertia, damping, and stiffness matrices. The corresponding kinematics are

$$\dot{\mathbf{v}}_d = \mathbf{a}_d, \quad (44)$$

$$\dot{\mathbf{p}}_d = \mathbf{T}(\boldsymbol{\theta}_d) \mathbf{v}_d, \quad (45)$$

with $\mathbf{a}_d = \dot{\mathbf{v}}_0 + \mathbf{a}_c$ from (43). Here $\mathbf{T}(\boldsymbol{\theta}_d)$ is the pose-rate-to-twist transfer matrix.

For the plasticity update, it is convenient to isolate the compliance-induced component of the motion, i.e., the deviation from the (time-varying) nominal reference $\mathbf{p}_0(t)$. Since $\mathbf{p}_0(t)$ may evolve along the planned trajectory, directly integrating $\mathbf{p}_d$ or $\mathbf{v}_d$ would undesirably accumulate nominal motion into the plastic state. We therefore define auxiliary states as errors with respect to the updated equilibrium $\mathbf{p}_0^*(t)$, so that only the interaction-driven deformation is captured by the plasticity dynamics:

$$\mathbf{v}_c \triangleq \boldsymbol{\xi}_v, \quad \mathbf{p}_c \triangleq \boldsymbol{\xi}_p, \quad (46)$$

with

$$\dot{\mathbf{v}}_c = \mathbf{a}_c, \quad (47)$$

$$\dot{\mathbf{p}}_c = \mathbf{T}(\boldsymbol{\theta}_d) \mathbf{v}_c. \quad (48)$$

2) Elasto-Plastic Deformation Update and Restoration:

The plasticity mechanism updates $\mathbf{p}_{\text{pl}}$ so that sustained interaction can shift the updated equilibrium (40). This prevents abrupt spring-back when an operator deliberately guides the MP to a new configuration (Fig. 7). The deformation dynamics is defined as

$$\dot{\mathbf{p}}_{\text{pl}} = \alpha \mathbf{v}_c + \beta \mathbf{p}_c - \gamma \mathbf{p}_{\text{pl}}, \quad (49)$$

where α and β are plasticity gains (enabled only when sustained interaction is detected), and γ is a restoration gain that enforces return towards the nominal equilibrium when

plasticity is disabled. Consistent with the supervisory logic, γ is activated only when plasticity is switched off:

$$\gamma = \begin{cases} \gamma_0, & \text{if } \alpha = 0 \text{ and } \beta = 0, \\ 0, & \text{otherwise,} \end{cases} \quad \gamma_0 > 0. \quad (50)$$

Thus, when α and β are set to zero, (49) reduces to $\dot{\mathbf{p}}_{\text{pl}} = -\gamma_0 \mathbf{p}_{\text{pl}}$, yielding an exponential decay and a smooth transition back to the nominal trajectory.

Accordingly, the equilibrium update can be expressed as

$$\mathbf{p}_{\text{pl}}(t) = \mathbf{p}_{\text{pl}}(0) + \int_0^t \dot{\mathbf{p}}_{\text{pl}}(s) ds, \quad (51)$$

$$\mathbf{p}_0^*(t) = \mathbf{p}_0(t) \oplus \mathbf{p}_{\text{pl}}(t). \quad (52)$$

The proposed update law (49) extends existing methods, such as the power-based EPRC [35], which commonly use binary triggers and direct velocity integration to induce equilibrium drift. In contrast, we formulate the equilibrium evolution as a continuous differential equation with three interpretable components:

- 1) **Velocity-driven plasticity** ($\alpha \mathbf{v}_c$): This term is proportional to the compliance velocity and provides an immediate, viscous-like response to dynamic interactions. Compared with purely position-driven updates, it improves responsiveness to rapidly varying external motions.
- 2) **Displacement-driven plasticity** ($\beta \xi_p$): This term depends on the accumulated displacement between the compliant pose and the updated equilibrium. It promotes convergence of the equilibrium during sustained quasi-static interaction, enabling the system to retain a deformed configuration even when the interaction velocity approaches zero.
- 3) **Restoration** ($-\gamma \mathbf{p}_{\text{pl}}$): This term enforces smooth recovery of the plastic state. While approaches such as EPRC may lead to persistent offsets that require manual resets or specific reverse motions [35], [47], the restoration term ensures convergence back to the nominal trajectory when interaction ceases.

In all experiments, the gains were set to $\alpha = 0.8$, $\beta = 0.2 \text{ s}^{-1}$, and $\gamma_0 = 0.5 \text{ s}^{-1}$. These values were selected by simulation-based tuning to obtain a stable compromise between three behaviours: a rapid initial compliant response to contact, a gradual plastic shift during sustained human guidance, and a smooth restoration toward the nominal reference after interaction termination.

3) Summary and Interface with the Overall Framework:

In summary, the elasto-plastic compliance controller maps the nominal task-space reference ($\mathbf{p}_0, \mathbf{v}_0, \dot{\mathbf{v}}_0$) and the estimated interaction wrench $\hat{\mathbf{w}}_{\text{int}}$ to the commanded reference ($\mathbf{p}_d, \mathbf{v}_d$) through admittance dynamics with the updated equilibrium (40). Under Strategy C, the admittance input is augmented by the boundary-repulsion term and, if needed, by the projection step to keep the commanded motion inside the wrench-feasible set (cf. (34)–(35)).

Compared with energy-triggered elasto-plastic strategies such as EPRC [35], the proposed law is better suited to dynamic PHRI because it updates the equilibrium without

waiting for energetic accumulation. The velocity-driven and displacement-driven plasticity terms allow the system to respond promptly to dynamic intent while remaining convergent under sustained quasi-static guidance.

During sustained interaction, (49) shifts the equilibrium to avoid abrupt snap-back, while (50) ensures smooth exponential recovery once plasticity is disabled. The resulting compliant reference is then passed to the tension-distribution and actuation layers (Fig. 4), thereby closing the loop with wrench estimation and contact classification.

C. Contacted-Cable Tension Regulation (Strategies B/C)

This subsection specifies how, once the supervisor grants release permission, the tension of the contacted cable is actually driven below its prescribed bound $\bar{\tau}_c$ without losing wrench feasibility. In contrast to prior cable-release methods that aim for $\tau_c \approx 0$ [27], [36], we regulate τ_c below an admissible bound, for two reasons: (i) driving τ_c to zero can severely shrink the feasible wrench set and is not safe to attempt in the majority of poses; and (ii) a contacted cable maintained at a bounded nonzero tension remains usable as a handle for collaborative guidance. The bound $\bar{\tau}_c$ is chosen so that the resulting contact force stays within human-safe limits. Also, the present framework no longer aims to keep the MP fixed after every cable contact. The elasto-plastic compliance law allows a limited displacement of the MP, and consequently of the contacted cable, away from the contact region.

The regulation has two coupled mechanisms, each mapped to one of the two hooks introduced in Section III-B: (1) the QP upper bound is scheduled to tighten the admissible tension on cable c ; (2) the joint-space controller for servomotor c switches to a reference-override mode that pays out cable length to enforce $\tau_c \leq \bar{\tau}_c$ against external pulling. The remaining cables keep the MP in a static equilibrium. The perception pipeline (Section IV) provides the required index c and timestamp t_{con} .

1) *Upper-bound scheduling and tension redistribution:* At the detected contact time t_{con} , the perception module provides the index c of the contacted cable and the filtered cable-tension vector $\boldsymbol{\tau}_{\text{mf}} \in \mathbb{R}^k$ ($k = 8$). The tension measured on the contacted cable is simply denoted by

$$\tau_{\text{mf},c} \triangleq [\boldsymbol{\tau}_{\text{mf}}]_c. \quad (53)$$

To avoid abrupt actuation changes and ensure a smooth transition into and out of contact management, we introduce a scalar scheduling function $s(t) \in [0, 1]$ with transition duration $T_r = 2 \text{ s}$. The admissible upper-tension bound of the contacted cable is then progressively reduced from its nominal value $[\boldsymbol{\tau}_{\text{max}}]_c$ to the management value $\bar{\tau}_c$, whereas the bounds of the remaining cables are left unchanged. Accordingly, the time-varying upper-bound vector is defined componentwise as

$$[\bar{\boldsymbol{\tau}}_{\text{max}}(t)]_i = \begin{cases} (1 - s(t))[\boldsymbol{\tau}_{\text{max}}]_c + s(t)\bar{\tau}_c, & i = c, \\ [\boldsymbol{\tau}_{\text{max}}]_i, & i \neq c. \end{cases} \quad (54)$$

At each control step during contact management, a feasible

tension distribution is recomputed by solving

$$\begin{aligned} \min_{\boldsymbol{\tau}} \quad & \|\boldsymbol{\tau}\|^2, \\ \text{subject to} \quad & \mathbf{W}_d \boldsymbol{\tau} = \mathbf{w}_{\text{req}}, \\ & \tau_{\min} \leq \tau \leq \bar{\tau}_{\max}(t), \end{aligned} \quad (55)$$

where $\mathbf{W}_d$ and $\mathbf{w}_{\text{req}}$ are defined as in Section III-B. This redistribution step preserves the required wrench at the MP while progressively tightening the admissible tension of the contacted cable.

2) *Cable reference override*: Although (55) enforces a reduced upper bound on the contacted cable in the tension-distribution stage, this alone does not guarantee that the actual cable tension will immediately fall below $\bar{\tau}_c$. Indeed, when a human pulls on the cable, the resulting external load increases the cable tension directly. Because the motor–winch transmission is effectively non-backdrivable, the excess tension cannot be relieved by passive backdriving alone, so the corresponding actuator must therefore be explicitly commanded to unwind the cable. For this reason, the reduction of the desired tension is complemented by a reference-override mode on motor c .

In this mode, actuator c is temporarily switched from nominal joint-position tracking to a motor-side tension-relief action that releases cable length until the measured tension drops below $\bar{\tau}_c$. The remaining actuators continue to track their nominal references so as to preserve the commanded MP motion.

We define the tension excess on the contacted cable as

$$\delta\tau_{f,c} \triangleq \tau_{\text{mf},c} - \bar{\tau}_c. \quad (56)$$

The corresponding incremental cable-release command is computed as

$$\Delta l_c = K_g T_s \text{sgn}_m(\delta\tau_{f,c}) + K_e T_s \delta\tau_{f,c}, \quad (57)$$

where T_s is the sampling period and K_g , K_e are scalar gains. The proportional term $K_e T_s \delta\tau_{f,c}$ provides a continuous correction for small-to-moderate tension excess, whereas the bounded high-gain term $K_g T_s \text{sgn}_m(\cdot)$ accelerates cable release for larger deviations without excessive sensitivity to small fluctuations. In implementation, we use a bounded modified sign function:

$$\text{sgn}_m(x) \triangleq \text{sat}\left(\frac{x}{\epsilon_\tau}\right), \quad \text{sat}(y) = \min\{1, \max\{-1, y\}\}, \quad (58)$$

with $\epsilon_\tau > 0$ defining the smoothing width.

To relate the proportional term to cable elasticity, K_e can be chosen consistently with Hooke's law, using the undeformed cable length l_u :

$$K_e = \frac{l_u}{ES}. \quad (59)$$

Finally, the incremental cable release is converted into a motor-angle correction through the drum radius r_d . The desired joint-angle vector is updated by modifying only the contacted actuator:

$$[\hat{\mathbf{q}}_d]_i = \begin{cases} [\hat{\mathbf{q}}_d]_c + \frac{\Delta l_c}{r_d}, & i = c, \\ [\hat{\mathbf{q}}_d]_i, & i \neq c. \end{cases} \quad (60)$$

3) *Interaction end detection and restoration*: Contact termination is detected using the incremental-release activity: when the closed-loop controller in (57) settles (i.e., Δl_c remains below a small threshold) for a prescribed hold time, the interaction is deemed to have ended. The rationale is that, under human–cable contact, external loading tends to keep $\tau_{\text{mf},c}$ above $\bar{\tau}_c$, which continuously triggers cable-length release. When the contact is removed, the tension falls below the bound and the release command vanishes apart from small measurement noise. Following [27], we declare the end of contact when $|\Delta l_c| < 0.1$ mm is satisfied continuously for 8 s. After termination, the scheduling function $s(t)$ is smoothly reset to restore the nominal upper bounds $\tau_{\max}$ and to return the contacted cable length toward its nominal value, thereby ensuring a continuous transition back to standard operation.

VI. RESULTS AND ANALYSIS

The experimental campaign on the [PlatformName] platform validates the framework at three scales, each mapped to one claim:

- **Claim I (interaction-flow, Subsection VI-A)**. The supervisor unifies transitions across interaction modes (safety-critical cable contact $\rightarrow$ collaborative cable use $\rightarrow$ direct MP manipulation) in a single continuous sequence.
- **Claim II (repeated-task trials, Subsection VI-B)**. The framework accurately reconstructs the human–robot interaction force from cable-tension sensing and robot modelling, without direct MP force sensing during operation. This reconstruction supports transparent, force-consistent, and non-amplifying elasto-plastic compliance, and is independently validated against a handle-mounted F/T sensor.
- **Claim III (user trials, Subsection VI-C)**. The framework is usable and collaboratively accessible to novices after minimal training.

The supplementary video material and supporting data are available through an anonymized link¹.

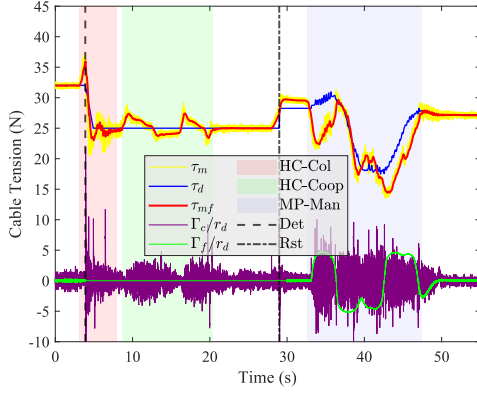
A. Interaction Flow: From Collision to Collaboration

Validation of Claim I: the supervisor handles a safety-critical human–cable contact, a cooperative human–cable interaction, and a direct human–MP manipulation as three phases of one continuous sequence, with automatic strategy switching and no operator-visible mode change.

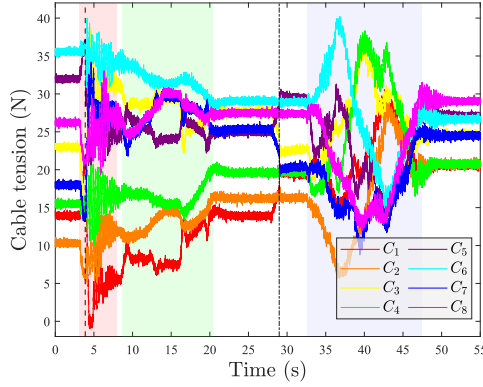
This subsection reports the closed-loop behaviour of the full perception–supervisor–compliance–tension-regulation pipeline over such a continuous sequence, representative of industrial and logistics settings where incidental contacts, intentional cooperative actions, and safety constraints must be handled seamlessly.

1) *Protocol and experimental conditions*: The robot operates with the full framework enabled (Fig. 4). A static reference is used so that the observed MP motion can be attributed directly to compliant response and operator action,

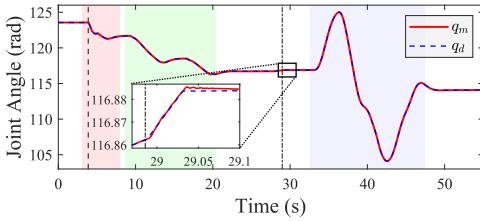
¹Additional anonymous experimental clips, organised by validation scenario and mapped to the corresponding manuscript sections, are available here.



(a) Contacted-cable signals (cable C_5): measured/desired tensions ($\tau_m, \tau_d, \tau_{mf}$) and normalised motor-torque terms ($\Gamma_c/r_d, \Gamma_f/r_d$). Timeline annotations use colour patches for interaction phases and vertical markers for key events (see caption).



(b) Measured tensions of all cables, τ_m , highlighting system-wide tension redistribution during the interaction flow. Vertical markers match those in (a).



(c) Motor tracking for the contacted cable: desired and measured motor angles (q_d, q_m) under mode switching (joint tracking versus tension tracking). Vertical markers match those in (a).

Fig. 8: Interaction-flow results over the 55 s analysis window (mapped to $t \in [0, 55]$ s). Background colour patches encode interaction phases: HC-Col (human–cable collision), HC-Coop (human–cable collaboration/handle use), and MP-Man (direct human–MP manipulation). Vertical markers indicate key events: Det, detection of the human–cable contact; and Rst, detected end of the human–cable contact and restoration of the contacted-cable tension bound.

rather than to a time-varying task trajectory. A representative dynamic trial is provided in the supplementary video.

The operator executes the following continuous sequence: (i) an unintentional collision with cable C_5 while not attending to the robot (HC-Col); (ii) an intentional pull on the same cable to bring the MP closer, using the cable as a handle (HC-Coop); (iii) a transition to direct MP manipulation by

applying forces to the MP (MP-Man).

This protocol is intended to illustrate a continuous transition from accidental collision to intentional collaboration. Accordingly, the operator does not wait for full restoration of the contacted-cable tension before stage (ii). Instead, after the accidental collision, the same cable is intentionally used as an interaction handle, so the contacted-cable tension is not expected to return to its nominal pre-contact value between stages (i) and (ii). For clarity, the plotted segment corresponds to an absolute time interval and is re-mapped to $t \in [0, 55]$ s in Fig. 8. The system response is analysed below by grouping the timeline into three stages.

2) *Collision detection and cable tension reduction*: The interaction begins with the MP holding a fixed nominal pose, with a nominal tension of approximately 32.60 N on cable C_5 . At $t = 3.17$ s, the operator collides with cable C_5 , producing a rapid increase in the measured contacted-cable tension, which peaks at 37.15 N. Compared with the later intentional cable-guidance phase, this collision phase shows a sharper transient and more pronounced oscillations in $\tau_{mf,5}$. The perception module (Section IV) confirms the human–cable collision at $t = 3.88$ s (marker Det in Fig. 8). Although the physical contact starts at $t = 3.17$ s, the tension error exceeds the threshold only after a 0.40 s rising phase. Once this threshold is crossed, the remaining algorithmic detection delay is 0.31 s, mainly due to the filtering window used to reduce false positives in the presence of sensor noise.

Upon detection and cable identification, the supervisor queries the RFSM associated with cable C_5 . Since the current MP pose is release-feasible and remains away from the RFSM boundary, the supervisor authorises Strategy B. Under Strategy B, the actuator associated with C_5 is explicitly commanded to regulate the contacted-cable tension rather than to maintain pure joint-position tracking; this is corroborated by the divergence between the measured motor angle q_m and the nominal desired angle q_d in Fig. 8c. The contacted-cable tension is progressively reduced toward the feasibility bound $\bar{\tau}_c = 25$ N and remains slightly below this limit, providing a compliant response while preserving feasible MP wrench generation.

3) *Seamless transition to collaboration*: Note that this collaboration phase starts before full restoration of the contacted-cable tension, since the same cable is intentionally kept in interaction after the accidental collision. After the accidental collision episode subsides ($t = 8.01$ s), the operator deliberately grasps the contacted cable at $t = 8.69$ s and pulls it to bring the MP closer (phase HC-Coop, ending at $t = 20.46$ s). This interaction modality is particularly convenient in constrained workspaces, as it allows the operator to reposition the MP without requiring close proximity to the platform itself. At $t = 28.99$ s, the contact-management module detects that the human–cable contact has ended and that the restoration conditions are satisfied (marker Rst).

Throughout this single-cable collaboration, feasibility is preserved at the system level, and the interaction strategy supervisor selects **Strategy B** (release cable C_5) during this phase. While C_5 is maintained at low tension to enable compliant interaction, the supervisor enforces wrench-balance

constraints by redistributing tensions across the remaining cables. In particular, the increment of C_1 tension during this phase is consistent with the change in load sharing required to accommodate the MP displacement induced by the pull on C_5 : as the platform moves, the cable geometry and the feasible tension allocation change, and some cables may unload while others take up compensating tension to satisfy equilibrium. Importantly, despite a modest increase in the contacted-cable tension during the pull, τ_5 remains below approximately 27 N in the representative trial of Fig. 8a. In the tension-distribution algorithm, $\bar{\tau}_c = 25$ N is imposed as the desired management bound. In closed-loop experiments, because the human directly loads the cable and the motor-side release is not instantaneous, the measured tension is regulated in a narrow neighbourhood of this target, with transient peaks up to 29 N over all reported trials. As shown in Fig. 8c, q_m converges back to q_d , indicating restoration of cable tension while maintaining payload support and thereby completing the transition from a safety-critical collision to normal operation within a single continuous scenario.

4) *MP manipulation*: At $t = 32.59$ s, the operator transfers the contact from the cable to the MP frame (phase MP-Man) and performs direct MP guidance until $t = 47.37$ s. During this phase, the elasto-plastic compliance controller generates MP motion from the estimated external wrench (Subsection IV-B). The system then smoothly restores the contacted-cable tension bound to its nominal value and switches motor control back to joint-tracking mode. Under nominal operation, the steady joint-angle tracking errors are typically below 0.001 rad.

B. Repeated-Task Trials: Quantification of Compliance

Validation of Claim II: during repeated physical guidance, the framework reconstructs the human–robot interaction force from cable-tension sensing and robot modelling without direct MP force sensing during operation. The reconstructed interaction force is dynamically consistent with an independent handle-mounted F/T measurement and supports transparent, force-consistent, non-amplifying MP compliance.

While the interaction flow demonstrates the functional integration of the framework, this subsection focuses on the quantitative behaviour of the elasto-plastic compliance controller under sustained physical guidance. Specifically, we characterise four complementary aspects: (i) *trajectory transparency and stabilising guidance*, by analysing the relationship between the measured MP motion and the adaptive plastic equilibrium; (ii) *dynamic validation of the model-based interaction-wrench estimate*, by directly comparing $\hat{\mathbf{w}}_{\text{int}}$ against measurements $\mathbf{w}_{\text{FT}}$ from a handle-mounted force/torque (F/T) sensor on the MP during dynamic human–MP interaction; (iii) *consistency between the model-based interaction-force estimate $\hat{\mathbf{f}}_{\text{int}}$ and the compliance-driving interaction proxy $\hat{\mathbf{f}}_{\text{imp}}$* used internally by the compliance loop, evaluated over the full repeated-task interval that may involve both cable and MP interaction modes; and (iv) *platform-level energetic behaviour*, by evaluating a power proxy at the interaction port together with the net power transmitted by the cable set at the MP. These analyses jointly quantify

whether the controller produces predictable compliant motion while remaining non-amplifying during repeated collaborative manipulation.

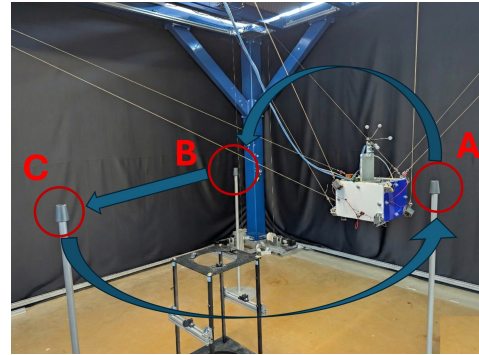


Fig. 9: Experimental setup for the repeated-task trials. The operator manually guides the 10.5 kg MP through a triangular via-point path defined by three calibrated markers (A, B, C) located at a height of 1.3 m.

1) *Protocol and experimental conditions*: The experimental setup is depicted in Fig. 9. Three spatial markers (labelled A, B, and C) are positioned at a height of 1.3 m, forming a horizontal triangle within the workspace. The markers are set to 1.30 m, i.e., near standing shoulder height for small operators and below shoulder height for most of the population, to avoid overhead postures [48]. The total mass of the MP is 10.5 kg.

The elasto-plastic compliance controller (Section V-B) is active with plasticity enabled, and the nominal reference $\mathbf{p}_0$ is held fixed at the centroid of the triangle. This configuration isolates the interaction behaviour: any displacement from the centroid is driven by physical user inputs and the induced plastic equilibrium update.

The operator is instructed to guide the MP through the sequence $A \rightarrow B \rightarrow C \rightarrow A$, repeating the loop five times continuously without pausing. The operator may interact either by grasping one cable at a time or by direct manipulation of the MP, and is free to choose the most convenient strategy (e.g., to minimise body repositioning). This continuous cyclic motion is designed to test the stability of the equilibrium update law and to accumulate interaction data for power analysis. The operator is not instructed to trace perfectly straight lines, so small lateral deviations are therefore expected and informative.

During the trials, the following variables are recorded:

- **Kinematics**: The ground-truth MP position $\mathbf{x}_m$ (MoCap), the commanded compliant reference $\mathbf{x}_d$, and the adaptive plastic equilibrium $\mathbf{x}_0^*$.
- **Dynamics**: The model-based interaction-force estimate $\hat{\mathbf{f}}_{\text{int}}$ and the impedance-consistent interaction proxy $\hat{\mathbf{f}}_{\text{imp}}$ used for compliance analysis.
- **Power flow (MP-level)**: The interaction-port power proxy P_{imp} and the cable-transmission power P_{ct} (cf. Section VI-B5).

2) *Trajectory & Transparency*: The kinematic performance of the proposed framework was evaluated through repeated navigation of the triangular path. Fig. 10 illustrates the planar trajectories, comparing the measured human–robot motion ($\mathbf{x}_m$) from MoCap, the adaptive plastic equilibrium ($\mathbf{x}_0^*$), and

the nominal linear references. Notably, a consistent lateral bias is visible during the $B \rightarrow C$ segment; as this path runs in close proximity to the supporting stand, the operator instinctively steers away from the structure, a behavioural adaptation that is clearly distinguished across the repeated cycles in Fig. 10.

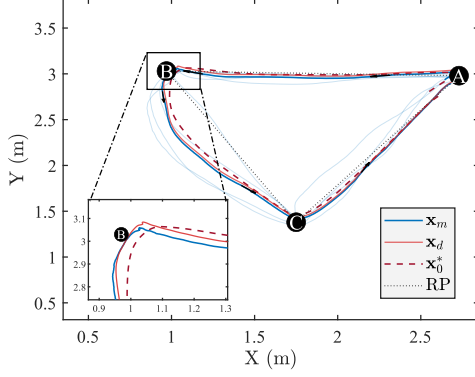


Fig. 10: Top-down (XY -plane) trajectories for the repeated task $A \rightarrow B \rightarrow C \rightarrow A$. The measured motion $\mathbf{x}_m$ from all repetitions is shown with low opacity, with one representative repetition highlighted in solid colour. The corresponding plastic equilibrium $\mathbf{x}_0^*$ and the desired reference $\mathbf{x}_d$ are also shown.

(i) *Position Tracking*: The tracking deviation is defined as $\mathbf{e} = \mathbf{x}_m - \mathbf{x}_d$. Across all five repetition cycles, the mean position errors (mm) are

$$X : 3.89 \pm 4.60, \quad Y : 1.15 \pm 4.30, \quad Z : -5.49 \pm 1.68,$$

where values are reported as mean $\pm$ standard deviation. These results indicate good repeatability in the horizontal plane despite physical human–robot contact, with standard deviations below 5 mm in both X and Y . In contrast, the Z direction exhibits a small dispersion but a consistent negative offset, suggesting a systematic vertical bias, which is likely attributable to cable elasticity and to the additional load of the communication cable (blue cable in Fig. 9).

(ii) *Transparency and Path Efficiency*:. To quantify the quality of physical interaction, two metrics were computed for each linear segment:

- 1) *RMS Lateral Deviation* (δ_{lat}): This metric measures how far the trajectory deviates laterally from the ideal straight line between the segment waypoints. It is defined as the root-mean-square of the orthogonal distance from each trajectory point to that line.

$$\delta_{\text{lat}} = \sqrt{\frac{1}{N} \sum_{i=1}^N \|\Pi_{\perp}(\mathbf{p}_i - \mathbf{a})\|^2}, \quad (61)$$

where $\mathbf{p}_i$ are XY positions along the segment, $\mathbf{a}$ is the segment start point, and $\Pi_{\perp}$ is the projection onto the direction orthogonal to the segment vector.

This serves as a proxy for transparency; lower deviation implies the system is not diverting the user from their intended linear path.

- 2) *Path Efficiency* (η): Defined as the ratio of the straight-line distance between the actual segment start and end

points ($D_{\text{endpoints}}$) to the total accumulated path length (L_{path}):

$$\eta = \frac{D_{\text{endpoints}}}{L_{\text{path}}}. \quad (62)$$

Each trial lasted 150 s and was sampled at 60 Hz, yielding 9000 measurements per repetition. Segment-wise metrics were computed for each repetition and are reported as mean $\pm$ SD over the five repetitions. Results for RMS lateral deviation are reported in Table II, and path efficiency in Table III.

TABLE II: RMS lateral deviation δ_{lat} (mm, mean $\pm$ SD) for the measured MP trajectory $\mathbf{x}_m$ and the plastic equilibrium trajectory $\mathbf{x}_0^*$.

Segment	Measured ($\mathbf{x}_m$)	Plastic eq. ($\mathbf{x}_0^*$)
$A \rightarrow B$	39.99 ± 26.75	21.79 ± 14.22
$B \rightarrow C$	138.17 ± 85.25	127.53 ± 61.81
$C \rightarrow A$	53.64 ± 21.38	45.77 ± 29.02

TABLE III: Path Efficiency (η) Analysis (mean $\pm$ SD).

Segment	Measured ($\mathbf{x}_m$)	Plastic eq. ($\mathbf{x}_0^*$)
$A \rightarrow B$	0.970 ± 0.023	0.998 ± 0.002
$B \rightarrow C$	0.907 ± 0.049	0.944 ± 0.039
$C \rightarrow A$	0.943 ± 0.044	0.993 ± 0.004

(iii) *Stabilizing Guidance of the Framework*: Across all segments, the plastic equilibrium $\mathbf{x}_0^*$ exhibits lower RMS lateral deviation than the measured motion $\mathbf{x}_m$ (with reductions up to $\sim 45\%$) and consistently higher path efficiency. These trends indicate that $\mathbf{x}_0^*$ captures the dominant interaction trend while attenuating local oscillations and small corrective motions, resulting in a smoother and straighter trajectory than the instantaneous hand-guided motion.

This behaviour is consistent with the elasto-plastic update law in (49). The equilibrium is not set to the instantaneous contact position. Instead, the displacement-driven term $\beta \xi_p$ integrates past deviations over time. As a result, the equilibrium update acts as a spatial low-pass filter that attenuates high-frequency chattering and small corrective motions during guidance, while preserving the dominant motion direction. Through the elastic restoring action, the compliant behaviour attracts the motion toward this filtered equilibrium, thereby assisting the operator in maintaining straighter segments without imposing a rigid trajectory.

3) *Wrench Estimator Validation*: To finish the repeated-task analysis, an additional dedicated experiment is reported here that validates the model-based interaction-wrench estimate $\hat{\mathbf{w}}_{\text{int}}$ (20) against direct measurements obtained from a six-axis F/T sensor rigidly integrated into a dedicated handle attached to the MP, as shown in Fig. 11a. The experiment is conducted during dynamic human–MP interaction, with the proposed framework fully active. The operator applies all interaction forces and moments through the instrumented handle, so that $\mathbf{w}_{\text{FT}}$ captures the full external wrench exerted on the MP.

By construction, this validation is restricted to human–MP interaction, since the handle-mounted F/T sensor cannot instru-

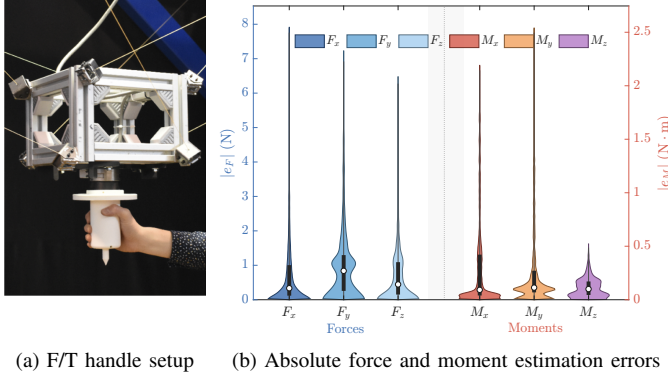


Fig. 11: Dynamic validation of the model-based interaction-wrench estimator using F/T-sensor measurements.

ment the cables. Let the measured wrench from the handle-mounted F/T sensor be decomposed as

$$\mathbf{w}_{FT} = \begin{bmatrix} \mathbf{f}_{FT} \\ \mathbf{m}_{FT} \end{bmatrix} \in \mathbb{R}^6, \quad (63)$$

where $\mathbf{f}_{FT} \in \mathbb{R}^3$ and $\mathbf{m}_{FT} \in \mathbb{R}^3$ are the measured interaction force and moment expressed at the MP geometric centre after the standard rigid-body transport from the sensor frame. The filtered model-based estimate from (20) is partitioned accordingly as

$$\hat{\mathbf{w}}_{int} = \begin{bmatrix} \hat{\mathbf{f}}_{int} \\ \hat{\mathbf{m}}_{int} \end{bmatrix} \in \mathbb{R}^6, \quad (64)$$

with $\hat{\mathbf{f}}_{int} \in \mathbb{R}^3$ and $\hat{\mathbf{m}}_{int} \in \mathbb{R}^3$ the estimated interaction force and moment at the MP. The force and moment estimation errors are then defined as

$$\mathbf{e}_F \triangleq \hat{\mathbf{f}}_{int} - \mathbf{f}_{FT}, \quad \mathbf{e}_M \triangleq \hat{\mathbf{m}}_{int} - \mathbf{m}_{FT}. \quad (65)$$

Figure 11b shows the empirical distributions of the signed componentwise force and moment errors over the dynamic interaction sequence. In the violin plots, the black circles and bars indicate the mean and standard deviation of each error component, respectively. The corresponding statistics, including the Euclidean norm computed at each time sample, are summarised in Table IV.

TABLE IV: External wrench estimation errors over the dynamic interaction sequence. Component errors are signed, whereas the norm is computed sample-wise. Values are mean $\pm$ standard deviation.

Error	x	y	z	$\ \cdot\ $
$\mathbf{e}_F$ [N]	0.005 ± 1.520	0.329 ± 1.387	0.377 ± 1.031	1.854 ± 1.454
$\mathbf{e}_M$ [N·m]	-0.001 ± 0.536	-0.074 ± 0.512	0.092 ± 0.110	0.524 ± 0.547

The signed per-axis means remain below 0.4 N for the force channels and below 0.1 N·m for the moment channels, indicating limited static bias. The norm errors, $\|\mathbf{e}_F\| = 1.85$ N and $\|\mathbf{e}_M\| = 0.52$ N·m on average, quantify the instantaneous combined error amplitude and remain small relative to the interaction loads (around 15 N) during guidance. The residual dispersion is consistent with cable-tension sensor noise, unmodelled cable/friction effects, and numerical differentiation

in the inverse dynamic model. These results support the use of $\hat{\mathbf{w}}_{int}$ as a reliable input for downstream compliance control.

4) *Force Consistency*: The repeated-task trials involve mixed interaction modes, including cable contact, direct MP contact, and transitions between them. Since no direct F/T ground truth is available for cable contacts, this subsection reports a consistency analysis between two reconstructed interaction-force signals: the model-based estimate $\hat{\mathbf{f}}_{int}$ from (19) and the compliance-driving proxy $\hat{\mathbf{f}}_{imp}$.

Because $\hat{\mathbf{f}}_{int}$ is obtained from the robot model and measured actuation quantities, $\hat{\mathbf{f}}_{imp}$ is reconstructed by back-driving the translational impedance dynamics using measured kinematics and the commanded compliant reference (and thus is sensitive to motion bandwidth, numerical differentiation, and time alignment). In particular, denoting the measured MP motion by $(\mathbf{x}_m, \dot{\mathbf{x}}_m, \ddot{\mathbf{x}}_m)$, the nominal velocity and acceleration by $(\dot{\mathbf{x}}_0, \ddot{\mathbf{x}}_0)$, and the adaptive plastic equilibrium by $\mathbf{x}_0^*$, the impedance-consistent estimate is reconstructed as

$$\hat{\mathbf{f}}_{imp} = \mathbf{M}(\ddot{\mathbf{x}}_m - \ddot{\mathbf{x}}_0) + \mathbf{B}(\dot{\mathbf{x}}_m - \dot{\mathbf{x}}_0) + \mathbf{K}(\mathbf{x}_m - \mathbf{x}_0^*), \quad (66)$$

where $\mathbf{M}$, $\mathbf{B}$, and $\mathbf{K}$ are the desired impedance matrices defined in (43). Accordingly, $\hat{\mathbf{f}}_{imp}$ is a *control-consistent* (impedance-shaped) force signal, whereas $\hat{\mathbf{f}}_{int}$ is an *estimation-consistent* force reflecting the inferred external loading. Because these signals originate from different constructs (impedance back-calculation versus model-based wrench estimation), the objective is not strict equality, but agreement in (a) temporal co-variation, (b) component-wise proportionality, and (c) 3D directional alignment.

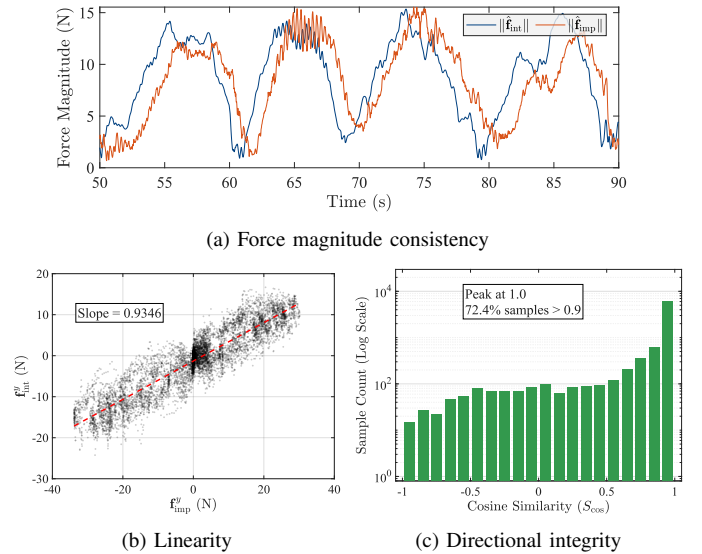


Fig. 12: Force consistency between the model-based estimate $\hat{\mathbf{f}}_{int}$ and the impedance-consistent proxy $\hat{\mathbf{f}}_{imp}$. (a) Magnitude time series (50–90 s). (b) Component-wise linearity along y (slope ≈ 0.93). (c) Cosine similarity during active interaction ($\|\hat{\mathbf{f}}_{int}\| > 2$ N), with 74.7% of samples satisfying $S_{cos} > 0.9$.

(i) Magnitude Co-variation and Proportionality:

Fig. 12a compares the force magnitudes $\|\hat{\mathbf{f}}_{int}\|$ and $\|\hat{\mathbf{f}}_{imp}\|$ over the repeated-task interval. The two traces evolve coherently, indicating that the compliance-driving force remains consistent

with the variations of the estimated interaction force. A slight delay of $\|\hat{\mathbf{f}}_{\text{imp}}\|$ with respect to $\|\hat{\mathbf{f}}_{\text{int}}\|$ can nevertheless be observed, which likely contributes to the small misalignment visible in Fig. 12c. This delay is expected because $\hat{\mathbf{f}}_{\text{imp}}$ is obtained from the MoCap through additional filtering and derivative estimation, which require forward-backward processing and therefore introduce extra latency. By contrast, $\hat{\mathbf{f}}_{\text{int}}$ is estimated directly from cable-tension measurements, which supports a more prompt force estimation.

(ii) *Linearity along the Principal Motion Axis:* To examine whether this consistency also holds component-wise, Fig. 12b reports a scatter comparison between $\hat{\mathbf{f}}_{\text{imp}}^y$ and $\hat{\mathbf{f}}_{\text{int}}^y$, where the Y direction corresponds to a principal in-plane motion component during the task. The data exhibit a clear linear trend, with an estimated slope of approximately 0.93. The complementary in-plane X direction shows a similar proportionality, with an estimated slope of about 0.92. These results indicate that the compliance loop preserves an approximately proportional mapping of the interaction intensity in the planar motion directions, even though the impedance-consistent force is not required to match the model-based wrench estimate exactly in magnitude.

(iii) *Directional Integrity via Cosine Similarity:* Beyond magnitude and component-wise proportionality, the most critical requirement for guidance and transparency is the preservation of the direction of the interaction force. We therefore compute the cosine similarity

$$S_{\cos}(t) = \frac{\hat{\mathbf{f}}_{\text{int}}(t)^\top \hat{\mathbf{f}}_{\text{imp}}(t)}{\|\hat{\mathbf{f}}_{\text{int}}(t)\| \|\hat{\mathbf{f}}_{\text{imp}}(t)\| + \varepsilon}, \quad (67)$$

with a small $\varepsilon > 0$ to avoid numerical singularities. The statistic is evaluated during active interaction phases only, using the criterion $\|\hat{\mathbf{f}}_{\text{int}}\| > 2\text{ N}$. As shown in Fig. 12c, the distribution of $S_{\cos}$ is strongly concentrated near +1, and 74.7% of the active samples satisfy $S_{\cos} > 0.9$. This confirms that $\hat{\mathbf{f}}_{\text{imp}}$ remains predominantly aligned with $\hat{\mathbf{f}}_{\text{int}}$ in 3D, ensuring that the compliant motion response is generated along directions consistent with the operator's wrench intent.

5) *Platform Dissipativity:* Non-amplifying behaviour at the MP level is prioritised to avoid runaway motion during physical guidance. In this framework, the intended benefit is achieved primarily through transparency and stabilisation (predictable compliant motion and bounded interaction), rather than through sustained energetic amplification at the platform. In the repeated-task trials, the MP weight is compensated by the model-based gravity term. Consequently, the operator mainly provides guidance forces, and the peak interaction-force magnitude remains below approximately 15 N (cf. Fig. 12a) while transporting a 10.5 kg platform.

(i) *Power definitions and sign conventions:* We characterise the energetic behaviour at the MP level by analysing instantaneous power flows associated with: (i) the interaction force signal used by the compliance loop, and (ii) the cable transmission acting on the MP. Because no dedicated force sensor is available at the human interface, the interaction-side power is interpreted as a power proxy rather than a ground-truth human power measurement.

Using the impedance-consistent force estimate $\hat{\mathbf{f}}_{\text{imp}}$ (Section VI-B4) and the measured MP velocity $\dot{\mathbf{x}}_m$, we define the interaction-port power proxy as

$$\mathbf{P}_{\text{imp}}(t) = \hat{\mathbf{f}}_{\text{imp}}(t)^\top \dot{\mathbf{x}}_m(t). \quad (68)$$

A positive value ($\mathbf{P}_{\text{imp}} > 0$) indicates that the interaction force proxy is aligned with the MP motion (net energy injection at this port), whereas $\mathbf{P}_{\text{imp}} < 0$ indicates net energy absorption.

At the MP level, the net power transmitted by the cable set is computed from the cable tensions τ_i and cable-length rates $\dot{l}_i$ as

$$\mathbf{P}_{\text{ct}}(t) = - \sum_{i=1}^k \tau_i(t) \dot{l}_i(t), \quad (69)$$

where k is the number of cables and $\dot{l}_i > 0$ denotes cable pay-out. With this convention, $\mathbf{P}_{\text{ct}} < 0$ corresponds to dissipative behaviour at the platform (energy extraction from the MP by the cable transmission), while $\mathbf{P}_{\text{ct}} > 0$ corresponds to transient net power delivery from the cable set to the MP (e.g., to satisfy motion commands or regulate tensions).

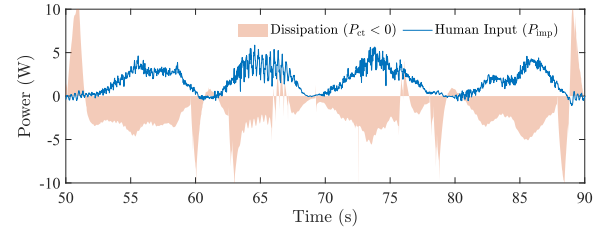


Fig. 13: Instantaneous MP-level powers: interaction-port proxy $\mathbf{P}_{\text{imp}}$ and cable-transmission power $\mathbf{P}_{\text{ct}}$

(ii) *Instantaneous power exchange at the platform:*

Fig. 13 compares the interaction-port power proxy $\mathbf{P}_{\text{imp}}$ and the cable-transmission power $\mathbf{P}_{\text{ct}}$. During many intervals where $\mathbf{P}_{\text{imp}} > 0$, the cable-transmission power is negative, i.e., $\mathbf{P}_{\text{ct}} < 0$, which corresponds to dissipative/braking behaviour of the cable transmission at the MP. However, short intervals with $\mathbf{P}_{\text{ct}} > 0$ are also observed, reflecting transient assistance or tension-regulation effects.

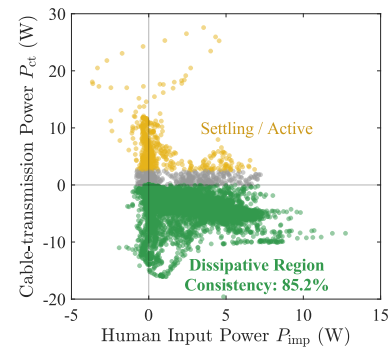


Fig. 14: Platform power quadrant map relating cable-transmission power $\mathbf{P}_{\text{ct}}$ to the interaction-port power proxy $\mathbf{P}_{\text{imp}}$. The concentration of samples in the region $\mathbf{P}_{\text{ct}} \leq 0$ indicates predominantly dissipative behaviour of the cable transmission at the MP during active guidance.

(iii) *Statistical view via a power quadrant map.*: Fig. 14 provides a statistical view of MP-level energetic behaviour by plotting P_{ct} against P_{imp} . Over the recorded samples, 85.2% lie in the region $P_{ct} \leq 0$, confirming that the cable transmission is predominantly dissipative at the platform during physical guidance. The remaining samples with $P_{ct} > 0$ correspond to short transient assistance phases (e.g., direction changes and tension regulation). Overall, this MP-level energetic indicator supports the conclusion that the proposed controller promotes stable, non-amplifying interaction behaviour during repeated collaborative manipulation.

C. User Trials: Collaborative Performance

Validation of Claim III: the framework is usable by novices after minimal training, with SUS scores above average and operator behaviour suggesting fluid transitions between cable and MP interaction.

1) *Participants:* Twenty-three participants were recruited for the experiments (22 males and 1 female; age range: 21–28 years). All were recruited from the same technical school to ensure a relatively homogeneous background, and none had prior experience in robotics research. All participants provided written informed consent prior to the study¹.

2) *Experimental Protocol:* The experimental task required each participant to manipulate the MP to trace a polyline trajectory through three waypoints (A, B, and C). The study comprised two phases (Experiments A and B) designed to assess *learnability* and *trajectory transparency* through objective metrics.

- **Experiment A (Naive Interaction):** During a pre-training familiarisation period, participants were allowed to manipulate the robot freely. They then performed 5 trials of the polyline-tracing task. Participants were free to interact using any preferred strategy (e.g., single-cable manipulation, multi-cable manipulation, direct contact with the MP, or simultaneous multiple contacts).
- **Training Phase:** After Exp. A, a researcher explained key system characteristics and demonstrated efficient interaction strategies. This three-minute training aimed to transfer minimal operational knowledge, enabling us to evaluate whether a short instruction period could improve collaborative performance and perceived interpretability. It also ensured that participants had enough familiarity with the system to meaningfully assess questionnaire items related to complexity, need for support, ease of learning, prior learning effort, and confidence.
- **Experiment B (Informed Interaction):** After training, participants performed 5 consecutive trials of the polyline-tracing task, applying the knowledge gained during the demonstration. Participants were free to use single or multiple contacts according to their preferences.

Experiments A and B were performed in a fixed sequence (Experiment A → training → Experiment B).

¹This work involved human subjects in its research. Approval of all ethical and experimental procedures and protocols was granted by [EthicsCommittee] under application no. [EthicsRef] (IRB no. [EthicsRef]) and performed in line with the regulatory qualification in non-interventional research.

3) *Data Collection and Analysis:* Each participant repeated Experiments A and B five times. This allowed us to assess within-subject repeatability and short-term learning. In both experiments, all participants started from the same initial position. All trials were video-recorded, and cable-tension and MP-position data were logged synchronously.

4) *Subjective Assessment:* At the conclusion of the session, participants completed a usability questionnaire. Responses were collected using a 5-point Likert scale (from “strongly disagree” to “strongly agree”). The survey consisted of the standard System Usability Scale (SUS) items [49], denoted U1–U10, and five custom items specific to the proposed collaborative system, denoted S1–S5. The full list of questions is provided in Appendix A. We acknowledge that the participant

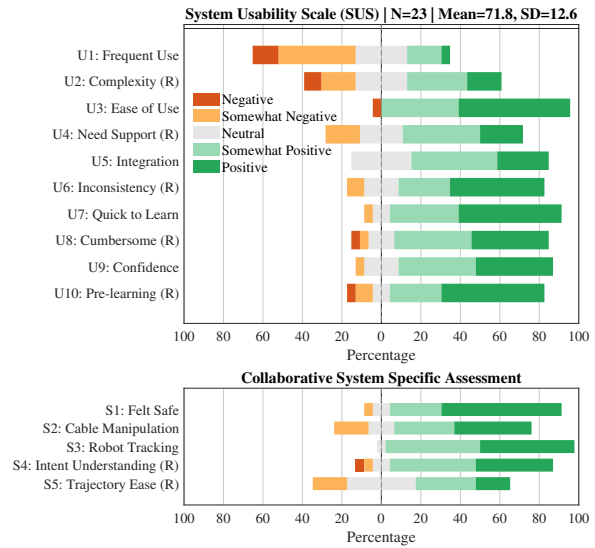


Fig. 15: Participants’ responses to the subjective questionnaires ($N = 23$). (R) denotes negatively phrased items, and responses are reverse-coded for visual consistency (green always indicates a positive user experience). Neutral responses are split equally around 0, and bar length shows percentages.

cohort was relatively homogeneous. This demographic profile is common in early-stage prototype studies but does not fully capture the diversity of industrial operators in terms of body size, professional experience, and fatigue levels encountered during extended shifts. Nevertheless, the results in Fig. 15 indicate a generally positive reception of the collaborative system, with strengths in perceived safety and robot tracking. Following the adjective ratings reported by Bangor *et al.* [50], the obtained SUS score of 71.8 falls within the “Good” category and exceeds the commonly cited average SUS score of 68 [49]. Using the percentile interpretation reported by Sauro [51], this score corresponds to approximately the 60th–65th percentile of SUS scores reported across usability studies. Notably, this score was obtained with an early-stage research prototype and minimal participant training, suggesting promising usability despite the absence of product-level integration.

Video observations indicate that participants often chose cable manipulation when the MP was out of reach, rather than walking to the MP. This behaviour is consistent with the questionnaire results, which show high agreement on ease of cable manipulation (S2) and robot tracking (S3). This perception of controllability is consistent with the repeated-task results

in Subsection VI-B, where the adaptive plastic equilibrium exhibited lower lateral deviation and higher path efficiency than the instantaneous measured trajectory. Participants also reported a strong sense of safety (S1), supporting the stability of the controller during physical human–robot collaboration. Although precise trajectory tracing remained a challenge (S5), the low demand for technical support (U4) and high confidence ratings (U9) indicate that a brief training phase enables novices to operate the system independently.

Although the supervisory framework of this paper is formulated for single-contact events, the user trials occasionally involved multiple simultaneous contacts in practice, for example when a participant touched the MP while still interacting with a cable, or when two cables were contacted at once. In these episodes, the overall behaviour remained stable, which is consistent with the predominantly non-amplifying MP-level behaviour reported in Subsection VI-B5. Qualitatively, when direct MP contact co-occurred with cable contact, the resulting motion tended to follow the operator’s intent at the MP; when two cables were contacted simultaneously, the observed motion reflected a combination of the two pulling directions. However, the present tension-management logic actively regulates only the first detected contacted cable, while additional simultaneous cable contacts are not yet explicitly managed. Extending contact perception and tension management to such multi-contact cases is therefore an important direction for future work.

VII. CONCLUSION

This paper introduced a feasibility-aware supervisory framework that unifies safety-critical and collaborative pHRI in CDPRs using only onboard cable-tension sensing, by coordinating compliant MP motion with contacted-cable tension regulation. The supervisor first verifies whether releasing the contacted cable is robustly feasible at the current pose. The elasto-plastic compliance law ensures continuity across reflex response, sustained guidance, and recovery. The contacted-cable regulation then keeps the contacted-cable tension below or around $\bar{\tau}_c$, rather than driving it to slackness, allowing the cable to serve as a controlled interaction channel.

The [PlatformName] validation supports three claims. The interaction-flow scenario showed unified switching across human-cable collision, Human-cable collaboration, and direct MP manipulation in one continuous sequence (Claim I). Repeated-task trials supported Claim II by showing accurate cable-tension-based wrench reconstruction, directionally consistent compliance forces, reduced lateral deviation through the plastic equilibrium, and predominantly dissipative MP-level behaviour. User trials with $N=23$ novices yielded an SUS score of 71.8 (“Good”), supporting practical accessibility despite the early-stage prototype (Claim III).

Future work will extend the framework beyond the single-contact cases validated in this paper. A first step is to assess multiple simultaneous contacts, such as multi-cable interaction or concurrent cable–MP contact, by extending the current supervisory logic to multi-cable constraints. A second direction is to enrich contact-source perception with

supplementary sensing, such as vision or local contact sensing, to better identify the contact location and posture and improve response selectivity. Finally, learning-based online adaptation of the supervisor’s gains will be investigated to reduce manual tuning and improve transparency and operator comfort while preserving safety and feasibility guarantees.

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APPENDIX

The following statements were rated by participants on a 5-point Likert scale.

System Usability Scale (SUS)

- **U1:** I think that I would like to use this system frequently.
- **U2:** I found this system unnecessarily complex.
- **U3:** I thought this system was easy to use.
- **U4:** I think that I would need assistance to be able to use this system.
- **U5:** I found the various functions in this system were well integrated.
- **U6:** I thought there was too much inconsistency in this system.
- **U7:** I would imagine that most people would learn to use this system very quickly.
- **U8:** I found this system very cumbersome/awkward to use.
- **U9:** I felt very confident using this system.
- **U10:** I needed to learn a lot of things before I could get going with this system.

System-Specific Questions

- **S1:** I felt safe during the cooperation with the robot.
- **S2:** I felt it was easy to use the cable manipulation.
- **S3:** The robot was following me easily.
- **S4:** The robot did not understand my intentions.
- **S5:** Following a pre-defined trajectory was difficult.